last time

register renaming motivation

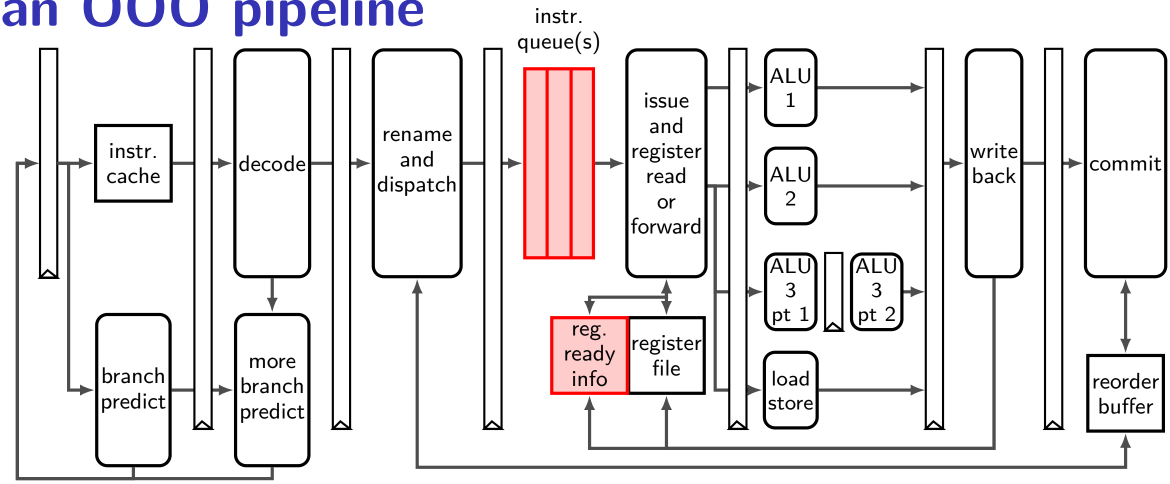
- different versions of registers
- out-of-order means multiple active versions

out-of-order pipeline

- in-order: fetch (w/ branch predict) + decode + rename + dispatch
- out-of-order: issue + execute + writeback
- in-order: commit

register renaming mechanics

an OOO pipeline



instruction queue and dispatch

instruction queue

#	*instruction*
1	addq %x01, %x05 → %x06
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
…	…

scoreboard

reg	*status*
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending
%x07	pending
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
…	…

execution unit
ALU 1
ALU 2

…

instruction queue and dispatch

instruction queue

#	instruction
1	`addq %x01, %x05 → %x06`
2	`addq %x02, %x06 → %x07`
3	`addq %x03, %x07 → %x08`
4	`cmpq %x04, %x08 → %x09.cc`
5	`jne %x09.cc, ...`
6	`addq %x01, %x08 → %x10`
7	`addq %x02, %x10 → %x11`
8	`addq %x03, %x11 → %x12`
9	`cmpq %x04, %x12 → %x13.cc`
...	...

scoreboard

reg	*status*
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending
%x07	pending
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
...	...

execution unit	*cycle# 1*	...
ALU 1	**1**	
ALU 2		

instruction queue and dispatch

instruction queue

#	*instruction*
1	addq %x01, %x05 → %x06
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
…	…

scoreboard

reg	*status*
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	pending
%x07	pending
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
…	…

execution unit	*cycle# 1*	…
ALU 1	1	
ALU 2		

instruction queue and dispatch

instruction queue

#	instruction
1	`addq %x01, %x05` → `%x06`
2	`addq %x02, %x06` → `%x07`
3	`addq %x03, %x07` → `%x08`
4	`cmpq %x04, %x08` → `%x09.cc`
5	`jne %x09.cc, ...`
6	`addq %x01, %x08` → `%x10`
7	`addq %x02, %x10` → `%x11`
8	`addq %x03, %x11` → `%x12`
9	`cmpq %x04, %x12` → `%x13.cc`
…	…

scoreboard

reg	status
`%x01`	ready
`%x02`	ready
`%x03`	ready
`%x04`	ready
`%x05`	ready
`%x06`	~~pending~~ ready
`%x07`	pending
`%x08`	pending
`%x09`	pending
`%x10`	pending
`%x11`	pending
`%x12`	pending
`%x13`	pending
…	…

execution unit	*cycle# 1*	…
ALU 1	1	
ALU 2	—	

instruction queue and dispatch

instruction queue

#	instruction
~~1~~	~~addq %x01, %x05 → %x06~~
2	addq %x02, %x06 → %x07
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
...	...

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	~~pending~~ ready
%x07	~~pending~~ ready
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
...	...

execution unit	cycle# 1	2	...
ALU 1	1	**2**	
ALU 2	—	—	

instruction queue and dispatch

instruction queue

#	instruction
~~1~~	~~addq %x01, %x05 → %x06~~
~~2~~	~~addq %x02, %x06 → %x07~~
3	addq %x03, %x07 → %x08
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
...	...

scoreboard

reg	*status*
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	~~pending~~ ready
%x07	~~pending~~ ready
%x08	~~pending~~ ready
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
...	...

execution unit	*cycle# 1*	*2*	*3*	...
ALU 1	1	2	**3**	
ALU 2	—	—	—	

instruction queue and dispatch

instruction queue

#	instruction
~~1~~	~~`addq %x01, %x05 → %x06`~~
~~2~~	~~`addq %x02, %x06 → %x07`~~
~~3~~	~~`addq %x03, %x07 → %x08`~~
4	`cmpq %x04, %x08 → %x09.cc`
5	`jne %x09.cc, ...`
6	`addq %x01, %x08 → %x10`
7	`addq %x02, %x10 → %x11`
8	`addq %x03, %x11 → %x12`
9	`cmpq %x04, %x12 → %x13.cc`
…	…

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	~~pending~~ ready
%x07	~~pending~~ ready
%x08	~~pending~~ ready
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
…	…

execution unit	*cycle# 1*	*2*	*3*	…
ALU 1	1	2	3	
ALU 2	—	—	—	

instruction queue and dispatch

instruction queue

#	instruction
~~1~~	~~addq %x01, %x05 → %x06~~
~~2~~	~~addq %x02, %x06 → %x07~~
~~3~~	~~addq %x03, %x07 → %x08~~
4	cmpq %x04, %x08 → %x09.cc
5	jne %x09.cc, ...
6	addq %x01, %x08 → %x10
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
...	...

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	~~pending~~ ready
%x07	~~pending~~ ready
%x08	~~pending~~ ready
%x09	~~pending~~ ready
%x10	~~pending~~ ready
%x11	pending
%x12	pending
%x13	pending
...	...

execution unit	*cycle# 1*	*2*	*3*	*4*	...
ALU 1	1	2	3	**4**	
ALU 2	—	—	—	**6**	

instruction queue and dispatch

instruction queue

#	*instruction*
~~1~~	~~addq %x01, %x05 → %x06~~
~~2~~	~~addq %x02, %x06 → %x07~~
~~3~~	~~addq %x03, %x07 → %x08~~
~~4~~	~~cmpq %x04, %x08 → %x09.cc~~
5	jne %x09.cc, ...
~~6~~	~~addq %x01, %x08 → %x10~~
7	addq %x02, %x10 → %x11
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
…	…

scoreboard

reg	*status*
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	~~pending~~ ready
%x07	~~pending~~ ready
%x08	~~pending~~ ready
%x09	~~pending~~ ready
%x10	~~pending~~ ready
%x11	pending
%x12	pending
%x13	pending
…	…

execution unit	*cycle# 1*	*2*	*3*	*4*	…
ALU 1	1	2	3	4	
ALU 2	—	—	—	6	

instruction queue and dispatch

instruction queue

#	instruction
~~1~~	~~addq %x01, %x05 → %x06~~
~~2~~	~~addq %x02, %x06 → %x07~~
~~3~~	~~addq %x03, %x07 → %x08~~
~~4~~	~~cmpq %x04, %x08 → %x09.cc~~
~~5~~	~~jne %x09.cc, ...~~
~~6~~	~~addq %x01, %x08 → %x10~~
~~7~~	~~addq %x02, %x10 → %x11~~
8	addq %x03, %x11 → %x12
9	cmpq %x04, %x12 → %x13.cc
…	…

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	~~pending~~ ready
%x07	~~pending~~ ready
%x08	~~pending~~ ready
%x09	~~pending~~ ready
%x10	~~pending~~ ready
%x11	~~pending~~ ready
%x12	pending
%x13	pending
…	…

execution unit	*cycle# 1*	*2*	*3*	*4*	*5*	…
ALU 1	1	2	3	4	**5**	
ALU 2	—	—	—	6	**7**	

instruction queue and dispatch

instruction queue

#	instruction
~~1~~	~~addq %x01, %x05 → %x06~~
~~2~~	~~addq %x02, %x06 → %x07~~
~~3~~	~~addq %x03, %x07 → %x08~~
~~4~~	~~cmpq %x04, %x08 → %x09.cc~~
~~5~~	~~jne %x09.cc, ...~~
~~6~~	~~addq %x01, %x08 → %x10~~
~~7~~	~~addq %x02, %x10 → %x11~~
~~8~~	~~addq %x03, %x11 → %x12~~
9	cmpq %x04, %x12 → %x13.cc
...	...

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	~~pending~~ ready
%x07	~~pending~~ ready
%x08	~~pending~~ ready
%x09	~~pending~~ ready
%x10	~~pending~~ ready
%x11	~~pending~~ ready
%x12	pending
%x13	pending
...	...

execution unit	*cycle#* 1	2	3	4	5	6	...
ALU 1	1	2	3	4	5	**8**	
ALU 2	—	—	—	6	7	—	

instruction queue and dispatch

instruction queue

#	instruction
~~1~~	~~`addq %x01, %x05 → %x06`~~
~~2~~	~~`addq %x02, %x06 → %x07`~~
~~3~~	~~`addq %x03, %x07 → %x08`~~
~~4~~	~~`cmpq %x04, %x08 → %x09.cc`~~
~~5~~	~~`jne %x09.cc, ...`~~
~~6~~	~~`addq %x01, %x08 → %x10`~~
~~7~~	~~`addq %x02, %x10 → %x11`~~
~~8~~	~~`addq %x03, %x11 → %x12`~~
~~9~~	~~`cmpq %x04, %x12 → %x13.cc`~~
…	…

scoreboard

reg	*status*
`%x01`	ready
`%x02`	ready
`%x03`	ready
`%x04`	ready
`%x05`	ready
`%x06`	~~pending~~ ready
`%x07`	~~pending~~ ready
`%x08`	~~pending~~ ready
`%x09`	~~pending~~ ready
`%x10`	~~pending~~ ready
`%x11`	~~pending~~ ready
`%x12`	~~pending~~ ready
`%x13`	pending
…	…

execution unit	*cycle# 1*	*2*	*3*	*4*	*5*	*6*	*7*	…
ALU 1	1	2	3	4	5	8	**9**	
ALU 2	—	—	—	6	7	—	**…**	

instruction queue and dispatch

instruction queue

#	instruction
~~1~~	~~`addq %x01, %x05 → %x06`~~
~~2~~	~~`addq %x02, %x06 → %x07`~~
~~3~~	~~`addq %x03, %x07 → %x08`~~
~~4~~	~~`cmpq %x04, %x08 → %x09.cc`~~
~~5~~	~~`jne %x09.cc, ...`~~
~~6~~	~~`addq %x01, %x08 → %x10`~~
~~7~~	~~`addq %x02, %x10 → %x11`~~
~~8~~	~~`addq %x03, %x11 → %x12`~~
~~9~~	~~`cmpq %x04, %x12 → %x13.cc`~~
…	…

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	~~pending~~ ready
%x07	~~pending~~ ready
%x08	~~pending~~ ready
%x09	~~pending~~ ready
%x10	~~pending~~ ready
%x11	~~pending~~ ready
%x12	~~pending~~ ready
%x13	~~pending~~ ready
…	…

execution unit	*cycle# 1*	*2*	*3*	*4*	*5*	*6*	*7*	…
ALU 1	1	2	3	4	5	8	9	
ALU 2	—	—	—	6	7	—	…	

instruction queue and dispatch

instruction queue

#	*instruction*
1	`movq (%x04) → %x06`
2	`movq (%x05) → %x07`
3	`addq %x01, %x02 → %x08`
4	`addq %x01, %x06 → %x09`
5	`addq %x01, %x07 → %x10`
…	…

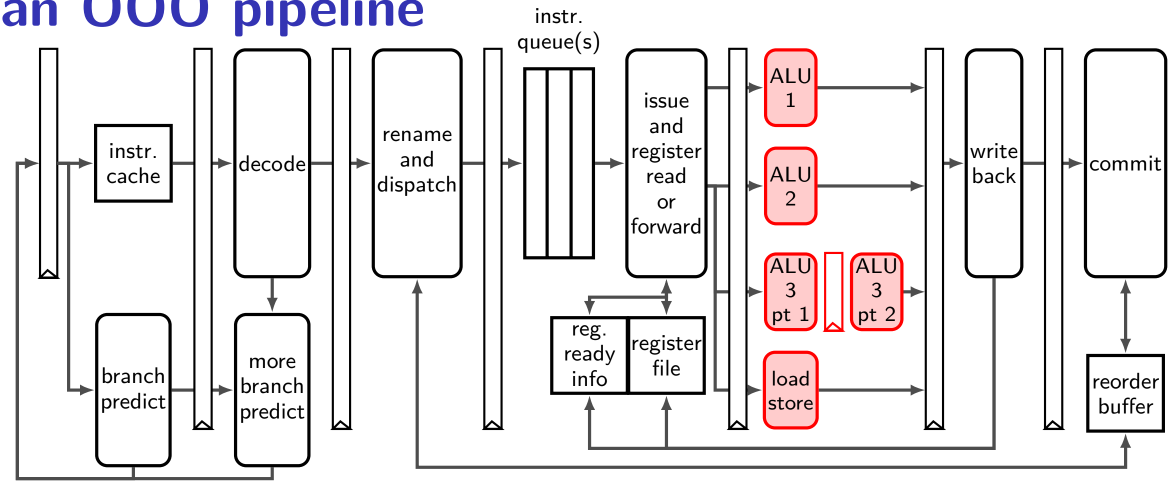
scoreboard

reg	*status*
%x01	ready
%x02	ready
%x03	ready
%x04	ready
%x05	ready
%x06	
%x07	
%x08	
%x09	
%x10	
…	…

execution unit	*cycle# 1*	*2*	*3*	*4*	*5*	*6*	*7*	…
ALU								
data cache								

assume
1 cycle/access

an OOO pipeline



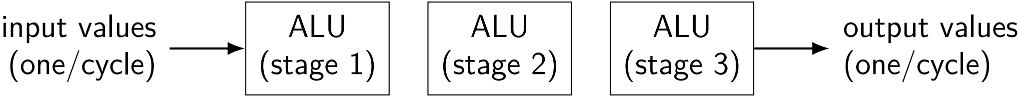
execution units AKA functional units (1)

where actual work of instruction is done

e.g. the actual ALU, or data cache

sometimes pipelined:

(here: 1 op/cycle; 3 cycle latency)



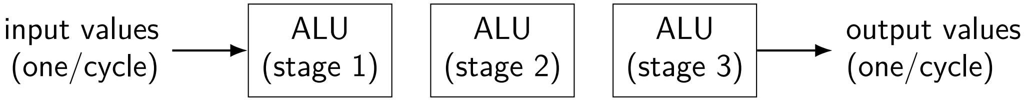
execution units AKA functional units (1)

where actual work of instruction is done

e.g. the actual ALU, or data cache

sometimes pipelined:

(here: 1 op/cycle; 3 cycle latency)



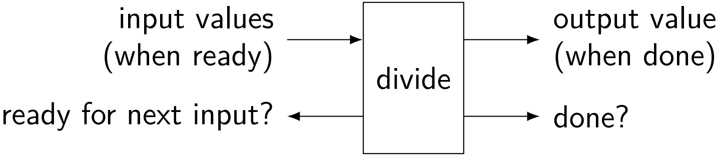
exercise: how long to compute $A \times (B \times (C \times D))$?

execution units AKA functional units (2)

where actual work of instruction is done

e.g. the actual ALU, or data cache

sometimes unpipelined:



instruction queue and dispatch (multicycle)

instruction queue

#	*instruction*
1	`add %x01, %x02 → %x03`
2	`imul %x04, %x05 → %x06`
3	`imul %x03, %x07 → %x08`
4	`cmp %x03, %x08 → %x09.cc`
5	`jle %x09.cc, ...`
6	`add %x01, %x03 → %x11`
7	`imul %x04, %x06 → %x12`
8	`imul %x03, %x08 → %x13`
9	`cmp %x11, %x13 → %x14.cc`
10	`jle %x14.cc, ...`
…	…

scoreboard

reg	*status*
%x01	ready
%x02	ready
%x03	pending
%x04	ready
%x05	ready
%x06	pending
%x07	ready
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
%x14	pending
…	… …

execution unit

ALU 1 (add, cmp, jxx)

ALU 2 (add, cmp, jxx)

ALU 3 (mul) start

ALU 3 (mul) end

instruction queue and dispatch (multicycle)

instruction queue

#	*instruction*
1	`add %x01, %x02 → %x03`
2	`imul %x04, %x05 → %x06`
3	`imul %x03, %x07 → %x08`
4	`cmp %x03, %x08 → %x09.cc`
5	`jle %x09.cc, ...`
6	`add %x01, %x03 → %x11`
7	`imul %x04, %x06 → %x12`
8	`imul %x03, %x08 → %x13`
9	`cmp %x11, %x13 → %x14.cc`
10	`jle %x14.cc, ...`
...	...

execution unit

ALU 1 (add, cmp, jxx)

ALU 2 (add, cmp, jxx)

ALU 3 (mul) start

ALU 3 (mul) end

scoreboard

reg	*status*
%x01	ready
%x02	ready
%x03	pending
%x04	ready
%x05	ready
%x06	pending
%x07	ready
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
%x14	pending
...	... ...

instruction queue and dispatch (multicycle)

scoreboard

instruction queue

#	*instruction*
1	`add %x01, %x02 → %x03`
2	`imul %x04, %x05 → %x06`
3	`imul %x03, %x07 → %x08`
4	`cmp %x03, %x08 → %x09.cc`
5	`jle %x09.cc, ...`
6	`add %x01, %x03 → %x11`
7	`imul %x04, %x06 → %x12`
8	`imul %x03, %x08 → %x13`
9	`cmp %x11, %x13 → %x14.cc`
10	`jle %x14.cc, ...`
…	…

reg	*status*
%x01	ready
%x02	ready
%x03	pending
%x04	ready
%x05	ready
%x06	pending
%x07	ready
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
%x14	pending
…	… …

execution unit	**cycle# 1**	
ALU 1 (add, cmp, jxx)	**1**	
ALU 2 (add, cmp, jxx)	**–**	
ALU 3 (mul) start	**2**	
ALU 3 (mul) end		**2**

instruction queue and dispatch (multicycle)

instruction queue

#	instruction
~~1~~	~~add %x01, %x02 → %x03~~
~~2~~	~~imul %x04, %x05 → %x06~~
3	imul %x03, %x07 → %x08
4	cmp %x03, %x08 → %x09.cc
5	jle %x09.cc, ...
6	add %x01, %x03 → %x11
7	imul %x04, %x06 → %x12
8	imul %x03, %x08 → %x13
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...
…	…

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	~~pending~~ ready
%x04	ready
%x05	ready
%x06	pending (still)
%x07	ready
%x08	pending
%x09	pending
%x10	pending
%x11	pending
%x12	pending
%x13	pending
%x14	pending
…	… …

execution unit	*cycle# 1*	**2**	
ALU 1 (add, cmp, jxx)	1	**6**	
ALU 2 (add, cmp, jxx)	–	**–**	
ALU 3 (mul) start	2	**3**	
ALU 3 (mul) end		2	**3**

instruction queue and dispatch (multicycle)

instruction queue

#	*instruction*
~~1~~	~~add %x01, %x02 → %x03~~
~~2~~	~~imul %x04, %x05 → %x06~~
~~3~~	~~imul %x03, %x07 → %x08~~
4	cmp %x03, %x08 → %x09.cc
5	jle %x09.cc, ...
~~6~~	~~add %x01, %x03 → %x11~~
7	imul %x04, %x06 → %x12
8	imul %x03, %x08 → %x13
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...
...	...

scoreboard

reg	*status*
%x01	ready
%x02	ready
%x03	~~pending~~ ready
%x04	ready
%x05	ready
%x06	~~pending~~ ready
%x07	ready
%x08	pending (still)
%x09	pending
%x10	pending
%x11	~~pending~~ ready
%x12	pending
%x13	pending
%x14	pending
...	... ...

execution unit	*cycle# 1*	*2*	**3**	
ALU 1 (add, cmp, jxx)	1	6	–	
ALU 2 (add, cmp, jxx)	–	–	–	
ALU 3 (mul) start	2	3	**7**	
ALU 3 (mul) end		2	3	**7**

instruction queue and dispatch (multicycle)

instruction queue

#	*instruction*
~~1~~	~~add %x01, %x02 → %x03~~
~~2~~	~~imul %x04, %x05 → %x06~~
~~3~~	~~imul %x03, %x07 → %x08~~
~~4~~	~~cmp %x03, %x08 → %x09.cc~~
5	jle %x09.cc, ...
~~6~~	~~add %x01, %x03 → %x11~~
~~7~~	~~imul %x04, %x06 → %x12~~
8	imul %x03, %x08 → %x13
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...
…	…

scoreboard

reg	*status*
%x01	ready
%x02	ready
%x03	~~pending~~ ready
%x04	ready
%x05	ready
%x06	~~pending~~ ready
%x07	ready
%x08	~~pending~~ ready
%x09	~~pending~~ ready
%x10	pending
%x11	~~pending~~ ready
%x12	pending (still)
%x13	pending
%x14	pending
…	… …

execution unit	*cycle# 1*	*2*	*3*	**4**	
ALU 1 (add, cmp, jxx)	1	6	–	**4**	
ALU 2 (add, cmp, jxx)	–	–	–	**–**	
ALU 3 (mul) start	2	3	7	**8**	
ALU 3 (mul) end		2	3	7	**8**

instruction queue and dispatch (multicycle)

instruction queue

#	instruction
~~1~~	~~add %x01, %x02 → %x03~~
~~2~~	~~imul %x04, %x05 → %x06~~
~~3~~	~~imul %x03, %x07 → %x08~~
~~4~~	~~cmp %x03, %x08 → %x09.cc~~
~~5~~	~~jle %x09.cc, ...~~
~~6~~	~~add %x01, %x03 → %x11~~
~~7~~	~~imul %x04, %x06 → %x12~~
8	imul %x03, %x08 → %x13
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...
...	...

scoreboard

reg	*status*
%x01	ready
%x02	ready
%x03	~~pending~~ ready
%x04	ready
%x05	ready
%x06	~~pending~~ ready
%x07	ready
%x08	~~pending~~ ready
%x09	~~pending~~ ready
%x10	pending
%x11	~~pending~~ ready
%x12	~~pending~~ ready
%x13	pending (still)
%x14	pending
...	... ...

execution unit	*cycle# 1*	*2*	*3*	*4*	**5**
ALU 1 (add, cmp, jxx)	1	6	–	4	**5**
ALU 2 (add, cmp, jxx)	–	–	–	–	–
ALU 3 (mul) start	2	3	7	8	–
ALU 3 (mul) end		2	3	7	8

instruction queue and dispatch (multicycle)

instruction queue

#	*instruction*
~~1~~	~~add %x01, %x02 → %x03~~
~~2~~	~~imul %x04, %x05 → %x06~~
~~3~~	~~imul %x03, %x07 → %x08~~
~~4~~	~~cmp %x03, %x08 → %x09.cc~~
~~5~~	~~jle %x09.cc, ...~~
~~6~~	~~add %x01, %x03 → %x11~~
~~7~~	~~imul %x04, %x06 → %x12~~
~~8~~	~~imul %x03, %x08 → %x13~~
9	cmp %x11, %x13 → %x14.cc
10	jle %x14.cc, ...
…	…

scoreboard

reg	*status*
%x01	ready
%x02	ready
%x03	~~pending~~ ready
%x04	ready
%x05	ready
%x06	~~pending~~ ready
%x07	ready
%x08	~~pending~~ ready
%x09	~~pending~~ ready
%x10	pending
%x11	~~pending~~ ready
%x12	~~pending~~ ready
%x13	~~pending~~ ready
%x14	pending
…	… …

execution unit	*cycle# 1*	*2*	*3*	*4*	*5*
ALU 1 (add, cmp, jxx)	1	6	–	4	5
ALU 2 (add, cmp, jxx)	–	–	–	–	–
ALU 3 (mul) start	2	3	7	8	–
ALU 3 (mul) end		2	3	7	8

instruction queue and dispatch (multicycle)

instruction queue

#	instruction
~~1~~	~~`add %x01, %x02 → %x03`~~
~~2~~	~~`imul %x04, %x05 → %x06`~~
~~3~~	~~`imul %x03, %x07 → %x08`~~
~~4~~	~~`cmp %x03, %x08 → %x09.cc`~~
~~5~~	~~`jle %x09.cc, ...`~~
~~6~~	~~`add %x01, %x03 → %x11`~~
~~7~~	~~`imul %x04, %x06 → %x12`~~
~~8~~	~~`imul %x03, %x08 → %x13`~~
~~9~~	~~`cmp %x11, %x13 → %x14.cc`~~
10	`jle %x14.cc, ...`
…	…

scoreboard

reg	status
%x01	ready
%x02	ready
%x03	~~pending~~ ready
%x04	ready
%x05	ready
%x06	~~pending~~ ready
%x07	ready
%x08	~~pending~~ ready
%x09	~~pending~~ ready
%x10	pending
%x11	~~pending~~ ready
%x12	~~pending~~ ready
%x13	~~pending~~ ready
%x14	~~pending~~ ready
…	… …

execution unit	*cycle# 1*	*2*	*3*	*4*	*5*	*6*
ALU 1 (add, cmp, jxx)	1	6	–	4	5	9
ALU 2 (add, cmp, jxx)	–	–	–	–	–	–
ALU 3 (mul) start	2	3	7	8	–	
ALU 3 (mul) end		2	3	7	8	

instruction queue and dispatch (multicycle)

instruction queue

#	*instruction*
~~1~~	~~add %x01, %x02 → %x03~~
~~2~~	~~imul %x04, %x05 → %x06~~
~~3~~	~~imul %x03, %x07 → %x08~~
~~4~~	~~cmp %x03, %x08 → %x09.cc~~
~~5~~	~~jle %x09.cc, ...~~
~~6~~	~~add %x01, %x03 → %x11~~
~~7~~	~~imul %x04, %x06 → %x12~~
~~8~~	~~imul %x03, %x08 → %x13~~
~~9~~	~~cmp %x11, %x13 → %x14.cc~~
~~10~~	~~jle %x14.cc, ...~~
…	…

scoreboard

reg	*status*
%x01	ready
%x02	ready
%x03	~~pending~~ ready
%x04	ready
%x05	ready
%x06	~~pending~~ ready
%x07	ready
%x08	~~pending~~ ready
%x09	~~pending~~ ready
%x10	pending
%x11	~~pending~~ ready
%x12	~~pending~~ ready
%x13	~~pending~~ ready
%x14	~~pending~~ ready
…	…

execution unit	*cycle# 1*	*2*	*3*	*4*	*5*	*6*	*7*
ALU 1 (add, cmp, jxx)	1	6	–	4	5	9	**10**
ALU 2 (add, cmp, jxx)	–	–	–	–	–	–	–
ALU 3 (mul) start	2	3	7	8	–		
ALU 3 (mul) end		2	3	7	8		

register renaming: missing pieces

what about "hidden" inputs like `%rsp`, condition codes?

one solution: translate to intructions with additional register parameters

- making %rsp explicit parameter
- turning hidden condition codes into operands!

bonus: can also translate complex instructions to simpler ones

```
popq %rax
```
→
```
movq (%rsp), %rax
addq $8, %rsp
```
→
```
movq (%x17) → %x19
addq $8, %x17 → %x18
```

```
cmpq %rax, %rbx
jle foo
```
→
```
cmpq %rax, %rbx, %CC
jle %CC, foo
```
→
```
cmpq %x01, %x04 → %x17
jle %x17, foo
```

OOO limitations

can't always find instructions to run

- plenty of instructions, but all depend on unfinished ones
- programmer can adjust program to help this

need to track all uncommitted instructions

- can only go so far ahead
- e.g. Intel Skylake: 224-entry reorder buffer, 168 physical registers

branch misprediction has a big cost (relative to pipelined)

- e.g. Intel Skylake: up to approx. 16 cycles (v. 2 for simple pipelined CPU)

OOO limitations

can't always find instructions to run

- plenty of instructions, but all depend on unfinished ones
- programmer can adjust program to help this

need to track all uncommitted instructions

- can only go so far ahead
- e.g. Intel Skylake: 224-entry reorder buffer, 168 physical registers

branch misprediction has a big cost (relative to pipelined)

- e.g. Intel Skylake: up to approx. 16 cycles (v. 2 for simple pipelined CPU)

some performance examples

```
example1:
    movq $10000000000, %rax
loop1:
    addq %rbx, %rcx
    decq %rax
    jge loop1
    ret
```

about 30B instructions

my desktop: approx 2.65 sec

```
example2:
    movq $10000000000, %rax
loop2:
    addq %rbx, %rcx
    addq %r8, %r9
    decq %rax
    jge loop2
    ret
```

about 40B instructions

my desktop: approx 2.65 sec

some performance examples

```
example1:
    movq $10000000000, %rax
loop1:
    addq %rbx, %rcx
    decq %rax
    jge loop1
    ret
```

about 30B instructions

my desktop: approx 2.65 sec

```
example2:
    movq $10000000000, %rax
loop2:
    addq %rbx, %rcx
    addq %r8, %r9
    decq %rax
    jge loop2
    ret
```

about 40B instructions

my desktop: approx 2.65 sec

importance of prediction

5-stage pipeline, predict not-taken versus always stall

hypothetical instruction mix

kind	portion	cycles (predict not-taken)	cycles (stall, no predict)
taken jXX	3%	3	3
non-taken jXX	5%	1	3
others	92%	1*	1*

* — ignoring data hazards

importance of prediction

5-stage pipeline, predict not-taken versus always stall

hypothetical instruction mix

kind	portion	cycles (predict not-taken)	cycles (stall, no predict)
taken jXX	3%	3	3
non-taken jXX	5%	1	3
others	92%	1*	1*

* — ignoring data hazards

prediction and OOO

deeper pipeline — much higher misprediction penalty

static branch prediction

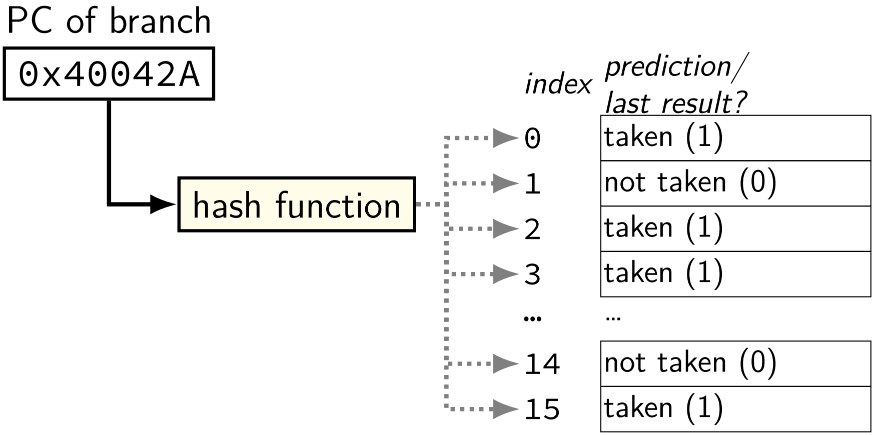
forward (target $>$ PC) not taken; backward taken

intuition: loops:

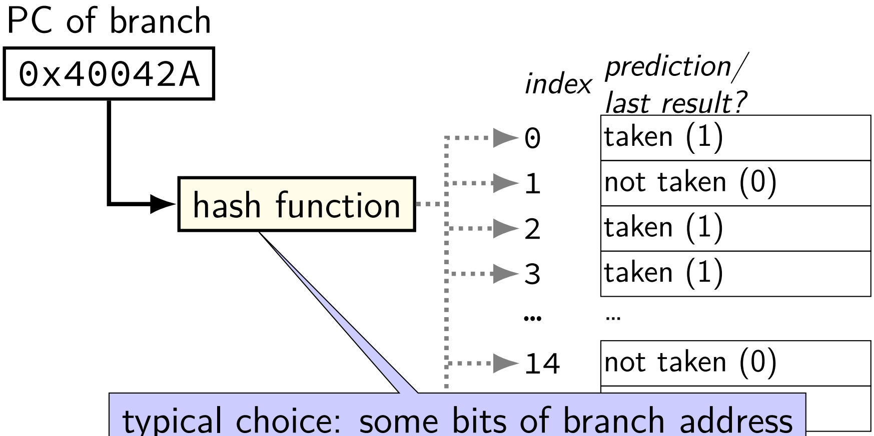
```
LOOP: ...
      ...
      je LOOP

LOOP: ...
      jne SKIP_LOOP
      ...
      jmp LOOP
SKIP_LOOP:
```

predict: repeat last

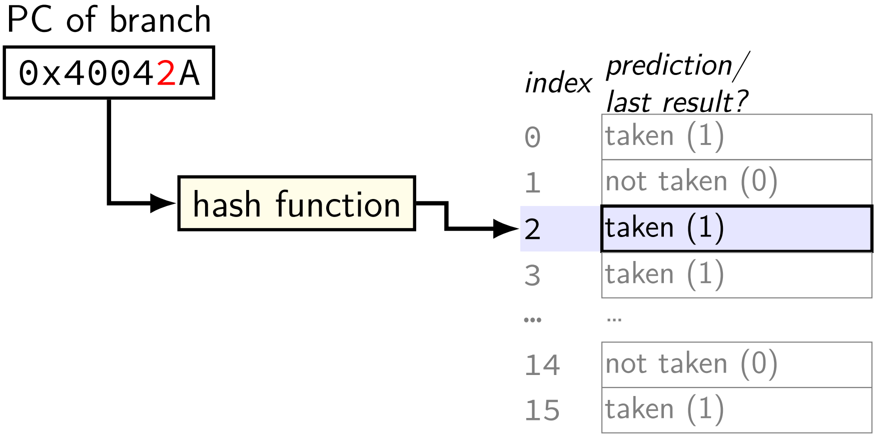


predict: repeat last

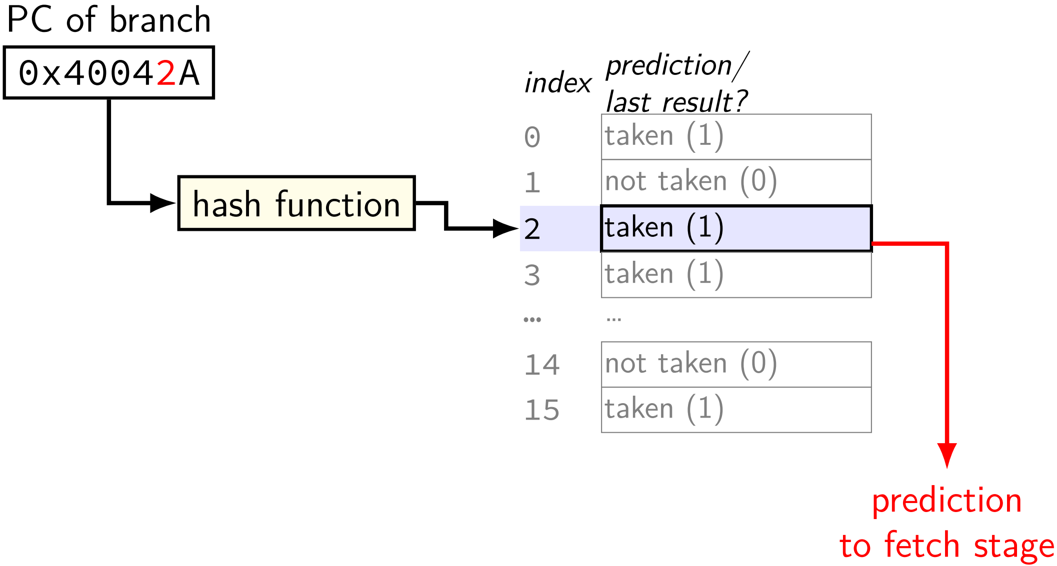


typical choice: some bits of branch address
for our example: will use bits 4-7

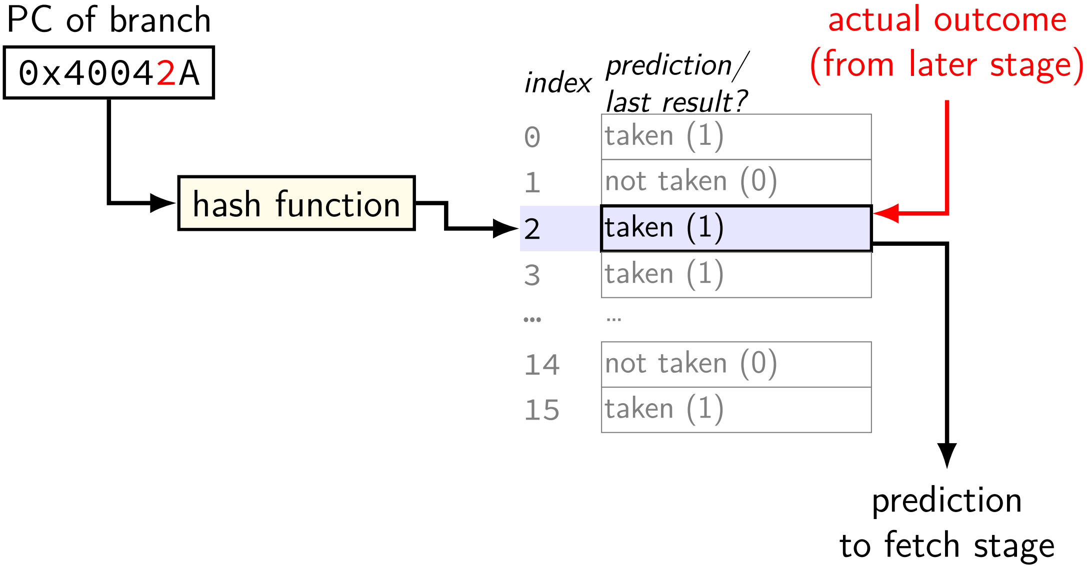
predict: repeat last



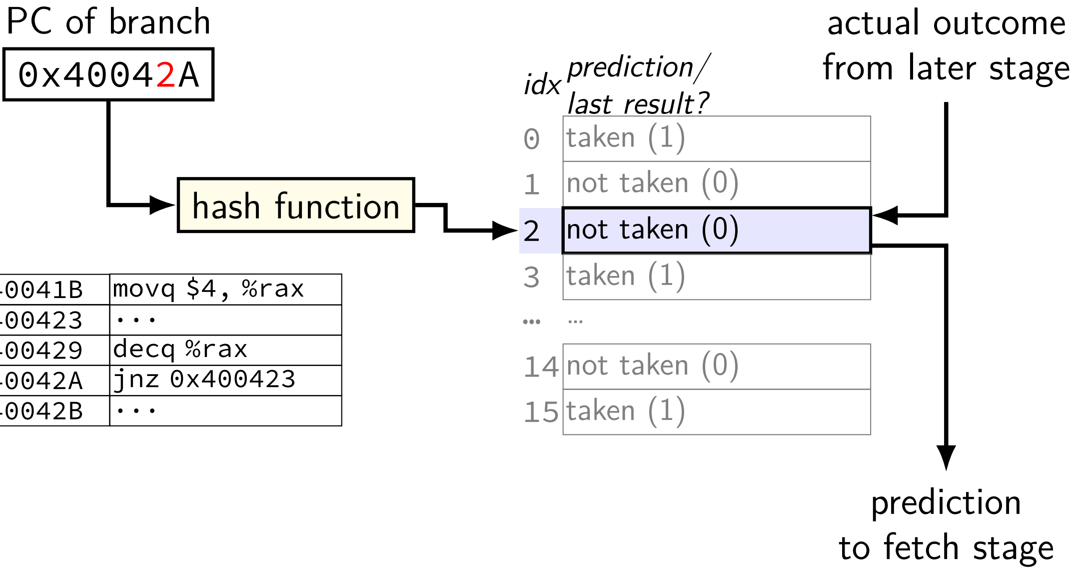
predict: repeat last



predict: repeat last

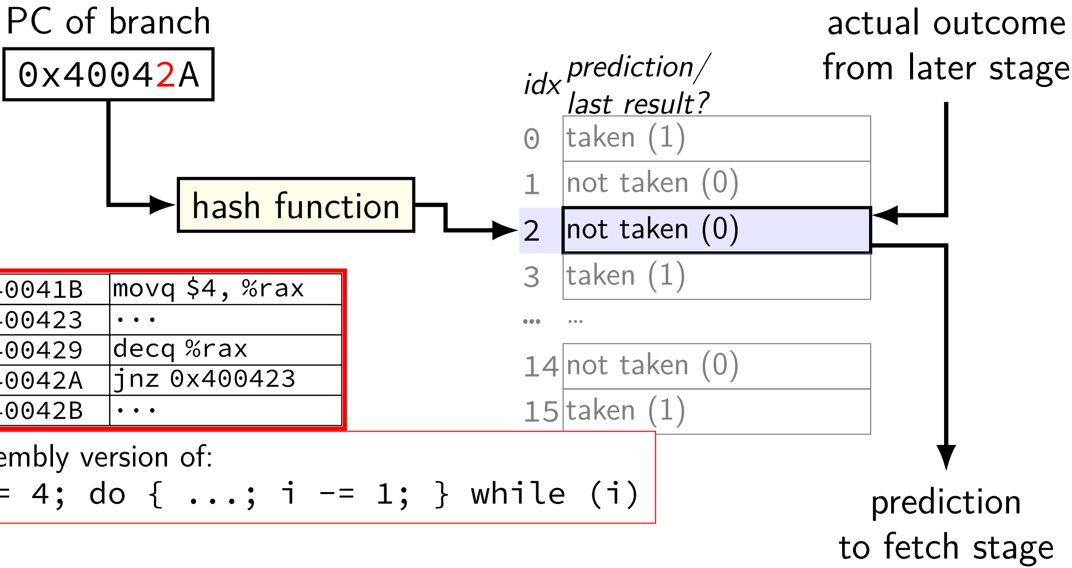


example



0x40041B	movq $4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jnz 0x400423
0x40042B	...

example

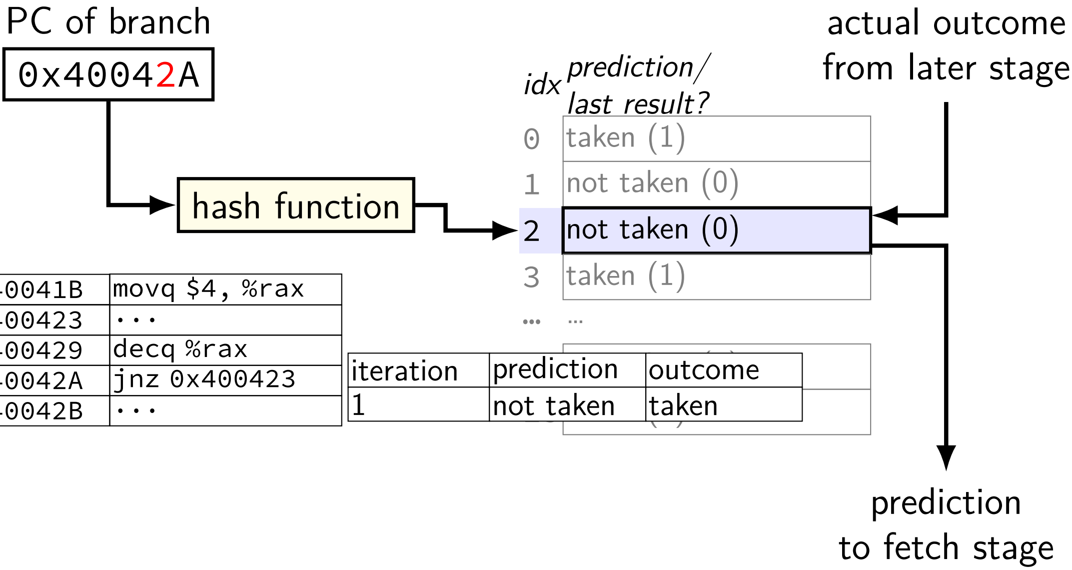


0x40041B	movq $4, %rax
0x400423	···
0x400429	decq %rax
0x40042A	jnz 0x400423
0x40042B	···

assembly version of:

```
i = 4; do { ...; i -= 1; } while (i)
```

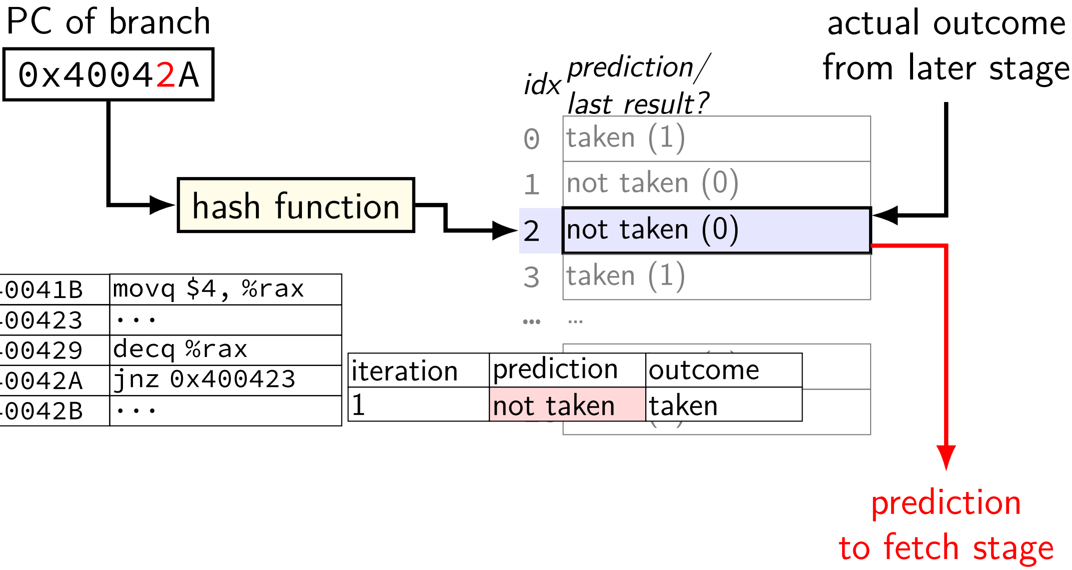
example



0x40041B	movq $4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jnz 0x400423
0x40042B	...

iteration	prediction	outcome
1	not taken	taken

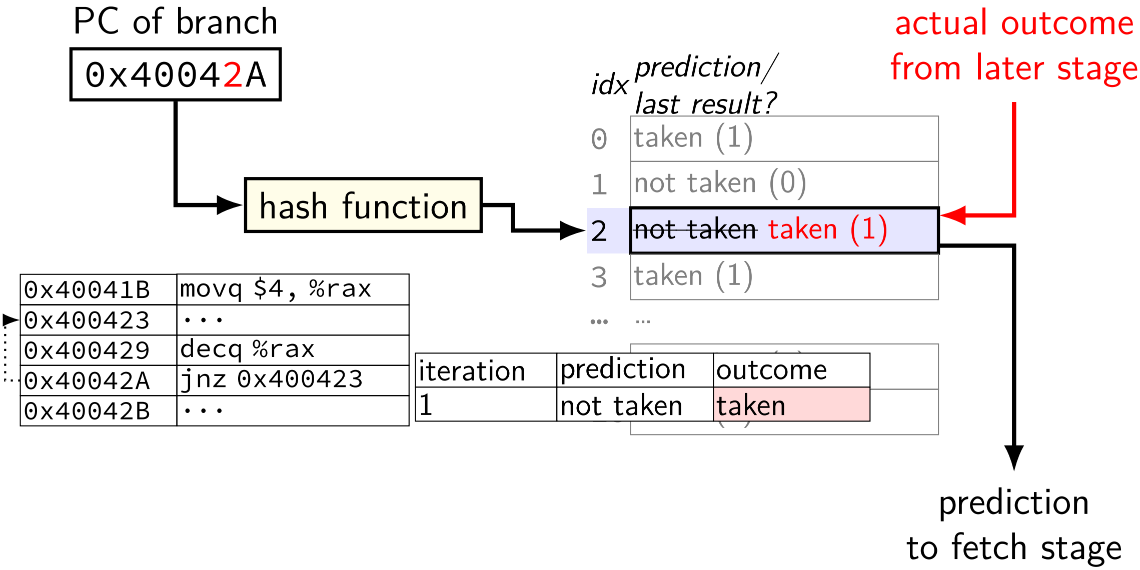
example



0x40041B	movq $4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jnz 0x400423
0x40042B	...

iteration	prediction	outcome
1	not taken	taken

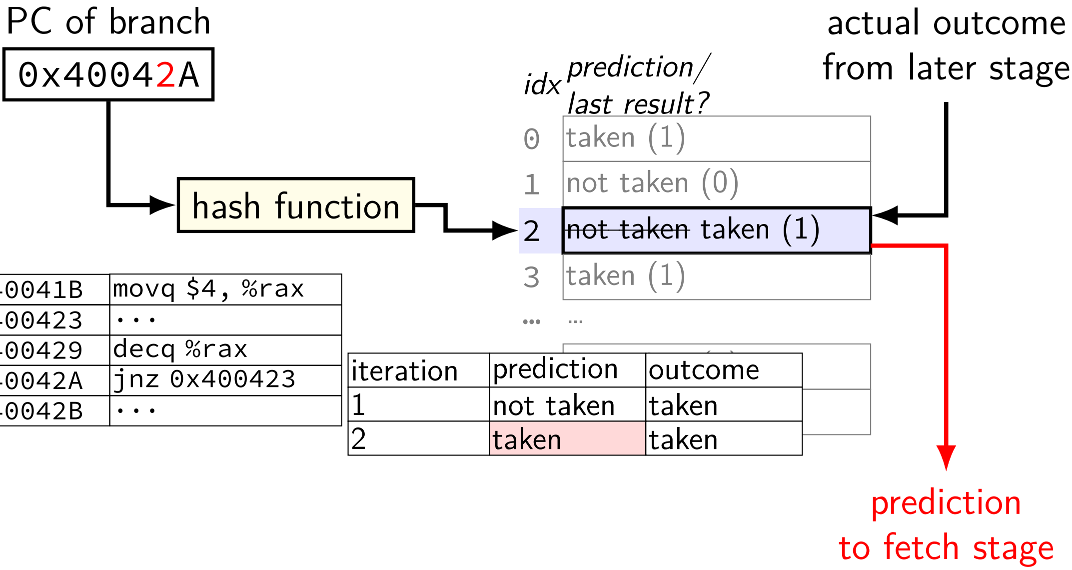
example



0x40041B	movq $4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jnz 0x400423
0x40042B	...

iteration	prediction	outcome
1	not taken	taken

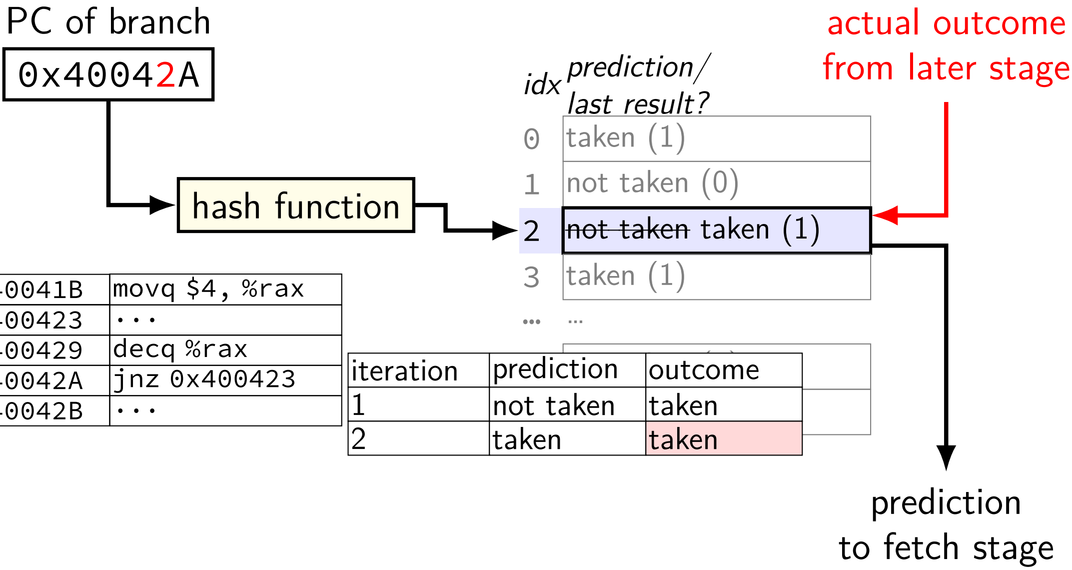
example



0x40041B	movq $4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jnz 0x400423
0x40042B	...

iteration	prediction	outcome
1	not taken	taken
2	taken	taken

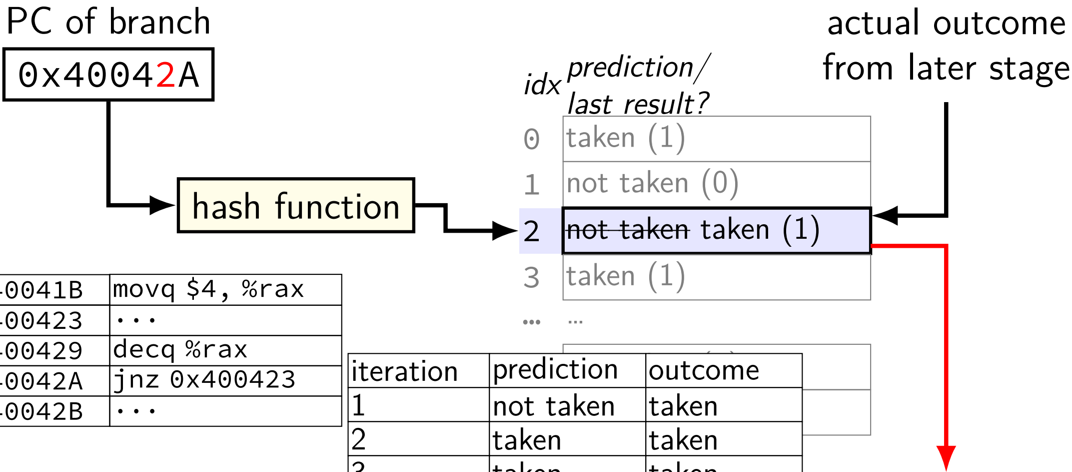
example



0x40041B	movq $4, %rax
0x400423	···
0x400429	decq %rax
0x40042A	jnz 0x400423
0x40042B	···

iteration	prediction	outcome
1	not taken	taken
2	taken	taken

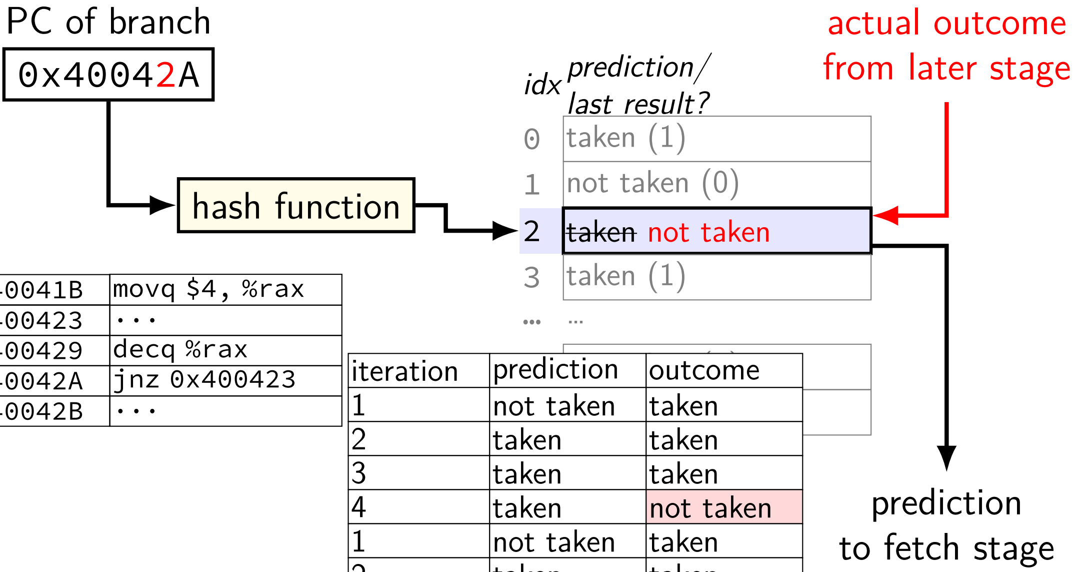
example



0x40041B	movq $4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jnz 0x400423
0x40042B	...

iteration	prediction	outcome
1	not taken	taken
2	taken	taken
3	taken	taken
4	taken	not taken
1	not taken	taken
2	taken	taken
...	...	...

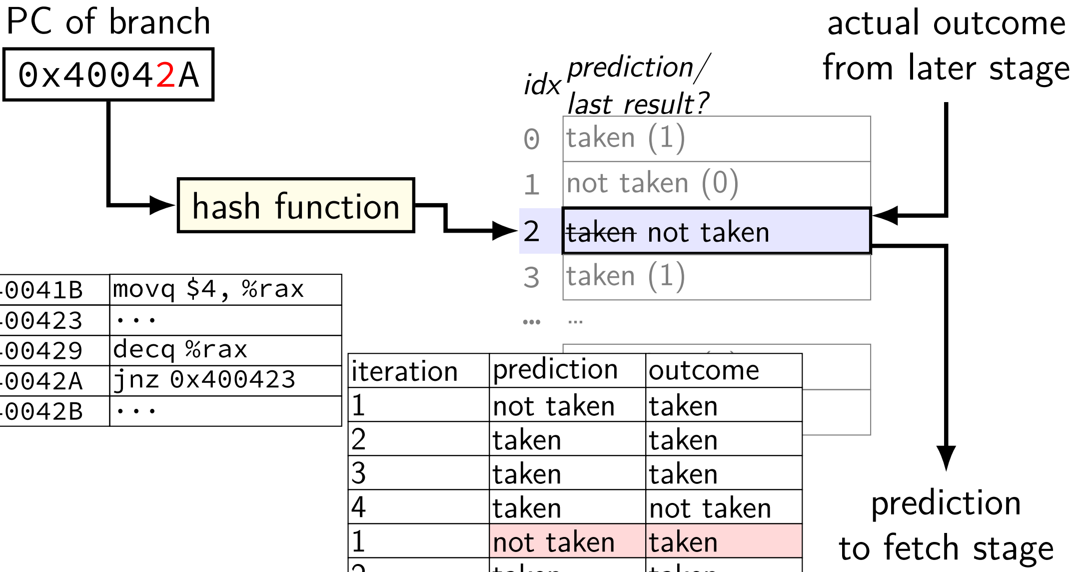
example



0x40041B	movq $4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jnz 0x400423
0x40042B	...

iteration	prediction	outcome
1	not taken	taken
2	taken	taken
3	taken	taken
4	taken	not taken
1	not taken	taken
2	taken	taken
...	...	...

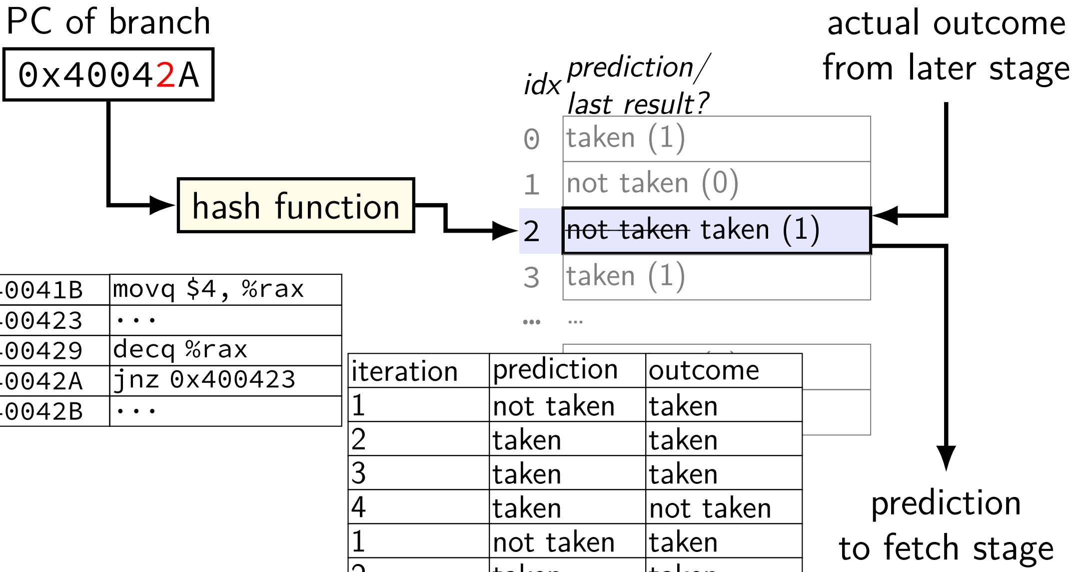
example



0x40041B	movq $4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jnz 0x400423
0x40042B	...

iteration	prediction	outcome
1	not taken	taken
2	taken	taken
3	taken	taken
4	taken	not taken
1	not taken	taken
2	taken	taken
...	...	...

example



0x40041B	movq $4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jnz 0x400423
0x40042B	...

iteration	prediction	outcome
1	not taken	taken
2	taken	taken
3	taken	taken
4	taken	not taken
1	not taken	taken
2	taken	taken
...	...	...

collisions?

two branches could have same hashed PC

nothing in table tells us about this

- versus direct-mapped cache: had *tag bits* to tell

is it worth it?

adding tag bits makes table *much* larger and/or slower

but does anything go wrong when there's a collision?

collision results

possibility 1: both branches usually taken

- no actual conflict — prediction is better(!)

possibility 2: both branches usually not taken

- no actual conflict — prediction is better(!)

possibility 3: one branch taken, one not taken

- performance probably worse

1-bit predictor for loops

- predicts first and last iteration wrong
  - example: branch to beginning — but same for branch from beginning to end
- everything else correct

exercise (pt 1)

use 1-bit predictor on this loop

- executed in outer loop (not shown) many, many times

what is the conditional jump misprediction rate for `i % 3 == 0`?

```
int i = 0;
while (true) {
  if (i % 3 == 0)
    goto next;
  ...
next:
  i += 1;
  if (i == 50)
    break;
}
```

exercise (pt 1)

use 1-bit predictor on this loop

executed in outer loop (not shown) many, many times

what is the conditional jump misprediction rate for `i % 3 == 0`?

```
int i = 0;
while (true) {
  if (i % 3 == 0)
    goto next;
  ...
next:
  i += 1;
  if (i == 50)
    break;
}
```

i =	branch	pred	outcome	correct?
0	mod 3	???	T	???
1	mod 3	T	F	no
2	mod 3	F	F	yes
3	mod 3	F	T	no
...	...	...	...	...

exercise (pt 1)

use 1-bit predictor on this loop

executed in outer loop (not shown) many, many times

what is the conditional jump misprediction rate for `i % 3 == 0`?

```
int i = 0;
while (true) {
  if (i % 3 == 0)
    goto next;
  ...
next:
  i += 1;
  if (i == 50)
    break;
}
```

i =	branch	pred	outcome	correct?
0	mod 3	???	T	???
1	mod 3	T	F	no
2	mod 3	F	F	yes
3	mod 3	F	T	no
…	…	…	…	…

exercise (pt 1)

use 1-bit predictor on this loop

- executed in outer loop (not shown) many, many times

what is the conditional jump misprediction rate for `i % 3 == 0`?

```
int i = 0;
while (true) {
  if (i % 3 == 0)
    goto next;
  ...
next:
  i += 1;
  if (i == 50)
    break;
}
```

i =	branch	pred	outcome	correct?
0	mod 3	???	T	???
1	mod 3	T	F	no
2	mod 3	F	F	yes
3	mod 3	F	T	no
…	…	…	…	…

exercise (pt 2)

use 1-bit predictor on this loop

- executed in outer loop (not shown) many, many times

what is the conditional jump misprediction rate for `i == 50`?

```
int i = 0;
while (true) {
  if (i % 3 == 0)
    goto next;
  ...
next:
  i += 1;
  if (i == 50)
    break;
}
```

exercise (full)

use 1-bit predictor on this loop

- executed in outer loop (not shown) many, many times

what is the conditional jump misprediction rate?

```
int i = 0;
while (true) {
  if (i % 3 == 0)
    goto next;
  ...
next:
  i += 1;
  if (i == 50)
    break;
}
```

exercise (full)

use 1-bit predictor on this loop

executed in outer loop (not shown) many, many times

what is the conditional jump misprediction rate?

```
int i = 0;
while (true) {
  if (i % 3 == 0)
    goto next;
  ...
next:
  i += 1;
  if (i == 50)
    break;
}
```

i =	branch	pred	outcome	correct?
0	mod 3	???	T	???
1	== 50	???	F	???
1	mod 3	T	F	—
2	== 50	F	F	✓
...	...	...	...	...

exercise (full)

use 1-bit predictor on this loop

executed in outer loop (not shown) many, many times

what is the conditional jump misprediction rate?

```
int i = 0;
while (true) {
  if (i % 3 == 0)
    goto next;
  ...
next:
  i += 1;
  if (i == 50)
    break;
}
```

i =	branch	pred	outcome	correct?
0	mod 3	???	T	???
1	== 50	???	F	???
1	mod 3	T	F	—
2	== 50	F	F	✓
...	...	...	...	...

branch target buffer

what if we can't decode LABEL from machine code for `jmp LABEL` or `jle LABEL` fast?

- will happen in more complex pipelines

what if we can't decode that there's a RET, CALL, etc. fast?

BTB: cache for branch targets

idx	valid	tag	ofst	type	target	(more info?)
0x00	1	0x400	5	Jxx	0x3FFFF3	…
0x01	1	0x401	C	JMP	0x401035	---
0x02	0	---	---	---	---	---
0x03	1	0x400	9	RET	---	…
…	…	…	…	…	…	…
0xFF	1	0x3FF	8	CALL	0x404033	…

valid	…
1	…
0	…
0	…
0	…
…	…
0	…

```
0x3FFFF3:  movq %rax, %rsi
0x3FFFF7:  pushq %rbx
0x3FFFF8:  call 0x404033
0x400001:  popq %rbx
0x400003:  cmpq %rbx, %rax
0x400005:  jle 0x3FFFF3
…          …
0x400031:  ret
…          …
```

BTB: cache for branch targets

idx	valid	tag	ofst	type	target	(more info?)
0x00	1	0x400	5	Jxx	0x3FFFF3	...
0x01	1	0x401	C	JMP	0x401035	---
0x02	0	---	---	---	---	---
0x03	1	0x400	9	RET	---	...
...	...	...	...	...	...	...
0xFF	1	0x3FF	8	CALL	0x404033	...

valid	...
1	...
0	...
0	...
0	...
...	...
0	...

```
0x3FFFF3:  movq %rax, %rsi
0x3FFFF7:  pushq %rbx
0x3FFFF8:  call 0x404033
0x400001:  popq %rbx
0x400003:  cmpq %rbx, %rax
0x400005:  jle 0x3FFFF3
...        ...
0x400031:  ret
...        ...
```

BTB: cache for branch targets

idx	valid	tag	ofst	type	target	(more info?)
0x00	1	0x400	5	Jxx	0x3FFFF3	…
0x01	1	0x401	C	JMP	0x401035	---
0x02	0	---	---	---	---	---
0x03	1	0x400	9	RET	---	…
…	…	…	…	…	…	…
0xFF	1	0x3FF	8	CALL	0x404033	…

valid	…
1	…
0	…
0	…
0	…
…	…
0	…

```
0x3FFFF3:  movq %rax, %rsi
0x3FFFF7:  pushq %rbx
0x3FFFF8:  call 0x404033
0x400001:  popq %rbx
0x400003:  cmpq %rbx, %rax
0x400005:  jle 0x3FFFF3
…          …
0x400031:  ret
…          …
```

indirect branch prediction

`jmp *%rax` or `jmp *(%rax, %rcx, 8)`

BTB can provide a prediction

but can do better with more context

example—predict based on other recent computed jumps

- good for polymophic method calls

table lookup with Hash(last few jmps)
instead of Hash(this jmp)

beyond 1-bit predictor

devote *more space* to storing history

main goal: rare exceptions don't immediately change prediction

example: branch taken 99% of the time

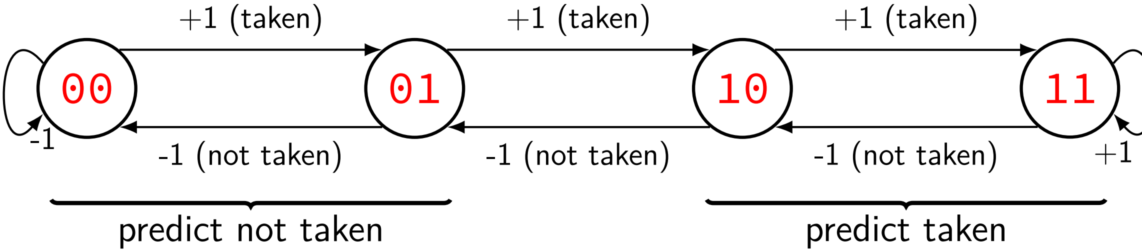
1-bit predictor: wrong about 2% of the time

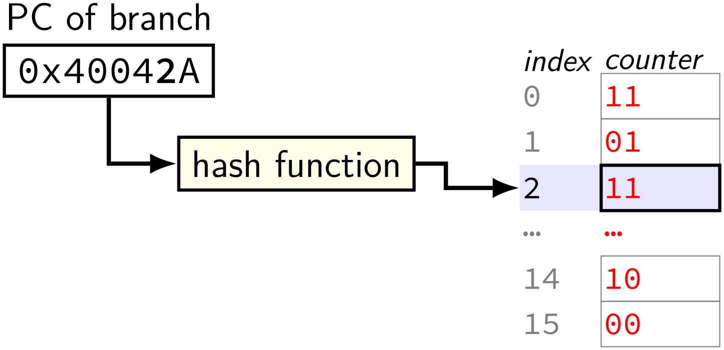
- 1% when branch not taken
- 1% of taken branches right after branch not taken

new predictor: wrong about 1% of the time

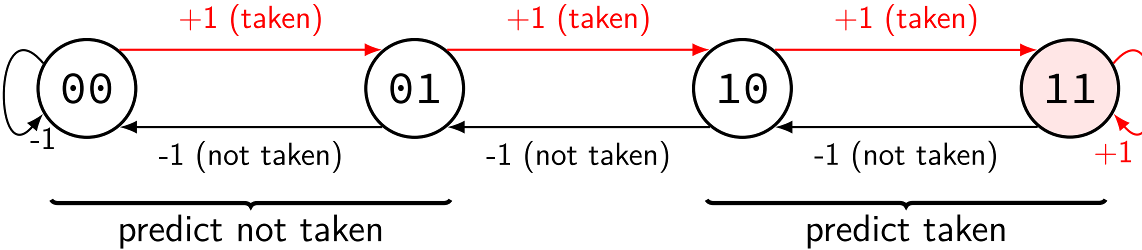
- 1% when branch not taken

2-bit saturating counter



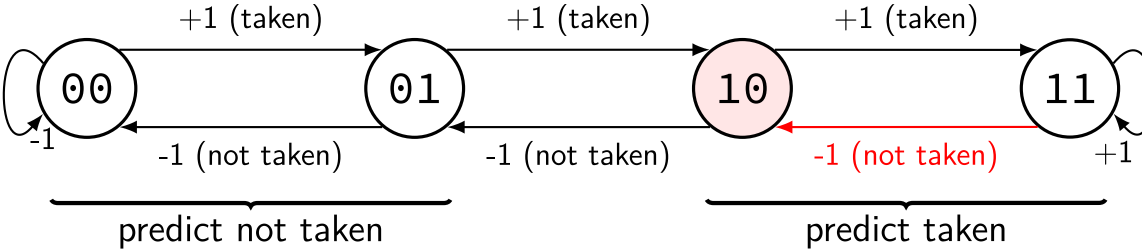


2-bit saturating counter



branch always taken:
value increases to 'strongest' taken value

2-bit saturating counter



branch almost always taken, then not taken once:
still predicted as taken

example

0x40041B	movq $4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jz 0x400423
0x40042B	...

iter.	table before	prediction	outcome	table after
1	01	not taken	taken	10
2	10	taken	taken	11
3	11	taken	taken	11
4	11	taken	not taken	10
1	10	taken	taken	11
2	11	taken	taken	11
3	11	taken	taken	11
4	11	taken	not taken	10
1	10	taken	taken	11
...	...	...	...	...

generalizing saturating counters

2-bit counter: ignore one exception to taken/not taken

3-bit counter: ignore more exceptions

`000` $\leftrightarrow$ `001` $\leftrightarrow$ `010` $\leftrightarrow$ `011` $\leftrightarrow$ `100` $\leftrightarrow$ `101` $\leftrightarrow$ `110` $\leftrightarrow$ `111`

`000-011`: not taken

`100-111`: taken

check_passphrase

```
int check_passphrase(const char *versus) {
    int i = 0;
    while (passphrase[i] == versus[i] &&
           passphrase[i]) {
        i += 1;
    }
    return (passphrase[i] == versus[i]);
}
```

number of iterations = number matching characters

leaks information about passphrase, oops!

exploiting check_passphrase (1)

guess	measured time
aaaa	100 ± 5
baaa	103 ± 4
caaa	102 ± 6
daaa	111 ± 5
eaaa	99 ± 6
faaa	101 ± 7
gaaa	104 ± 4
…	…

exploiting check_passphrase (2)

guess	measured time
daaa	102 ± 5
dbaa	99 ± 4
dcaa	104 ± 4
ddaa	100 ± 6
deaa	102 ± 4
dfaa	109 ± 7
dgaa	103 ± 4
…	…

timing and cryptography

lots of asymmetric cryptography uses big-integer math

example: multiplying 500+ bit numbers together

how do you implement that?

big integer multiplcation

say we have two 64-bit integers x, y

and want to 128-bit product, but our multiply instruction only does 64-bit products

one way to multiply:

divide x, y into 32-bit parts: $x = x_1 \cdot 2^{32} + x_0$ and $y = y_1 \cdot 2^{32} + y_0$

then $xy = x_1 y_1 2^{64} + x_1 y_0 \cdot 2^{32} + x_0 y_1 \cdot 2^{32} + x_0 y_0$

big integer multiplcation

say we have two 64-bit integers x, y

and want to 128-bit product, but our multiply instruction only does 64-bit products

one way to multiply:

divide x, y into 32-bit parts: $x = x_1 \cdot 2^{32} + x_0$ and $y = y_1 \cdot 2^{32} + y_0$

then $xy = x_1 y_1 2^{64} + x_1 y_0 \cdot 2^{32} + x_0 y_1 \cdot 2^{32} + x_0 y_0$

can extend this idea to arbitrarily large numbers

number of smaller multiplies depends on size of numbers!

big integers and cryptography

naive multiplication idea:

- number of steps depends on size of numbers

problem: sometimes the value of the number is a secret

- e.g. part of the private key

oops! revealed through timing

big integer timing attacks in practice (1)

early versions of OpenSSL (TLS implementation)had timing attack

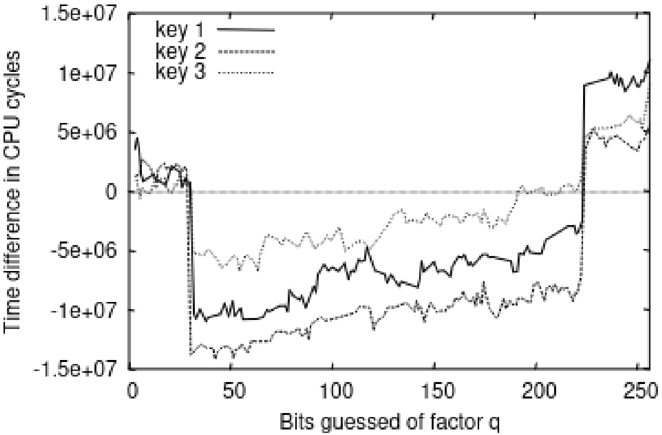
- Brumley and Boneh, "Remote Timing Attacks are Practical" (Usenix Security '03)

attacker could figure out bits of private key from timing

why? variable-time mulitplication and modulus operations

- got faster/slower depending on how input was related to private key

big integer timing attacks in practice (2)



(a) The zero-one gap $T_g - T_{g_{hi}}$ indicates that we can distinguish between bits that are 0 and 1 of the RSA factor q for 3 different randomly-generated keys. For clarity, bits of q that are 1 are omitted, as the x-axis can be used for reference for this case.

Figure 3a from Brumley and Boneh, "Remote Timing Attacks are Practical"

browsers and website leakage

web browsers run code from untrusted webpages

one goal: can't tell what other webpages you visit

some webpage leakage (1)

…as you can see here, here, and here …

convenient feature 1: browser marks visited links

```
<script>
var the_color = window.getComputedStyle(
    document.querySelector('a[href=~"foo.com"]')
).color
if (the_color == ...) { ... }
</script>
```

convenient feature 2: scripts can query current color of something

some webpage leakage (1)

…as you can see here, here, and here …

convenient feature 1: browser marks visited links

```
<script>
var the_color = window.getComputedStyle(
    document.querySelector('a[href=~"foo.com"]')
).color
if (the_color == ...) { ... }
</script>
```

~~convenient feature 2: scripts can query current color of something~~

fix 1: getComputedStyle lies about the color

fix 2: limited styling options for visited links

some webpage leakage (2)

one idea: script in webpage times loop that writes big array

variation in timing depends on other things running on machine

some webpage leakage (2)

one idea: script in webpage times loop that writes big array

variation in timing depends on other things running on machine

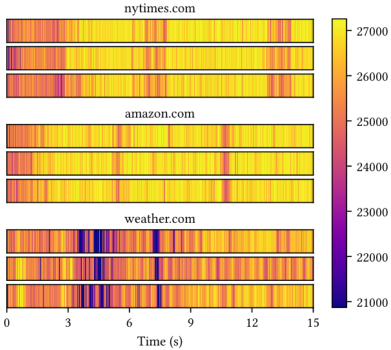


Figure 3: Example loop-counting traces collected over 15 seconds. Darker shades indicate smaller counter values and lower instruction throughput.

turns out, other webpages create distinct "signatures"

Figure from Cook et al, "There's Always a Bigger Fish: Clarifying Analysis of Machine-Learning-Assisted Side-Channel Attack" (ISCA '22)

inferring cache accesses (1)

suppose I time accesses to array of chars:

- reading array[0]: 3 cycles
- reading array[64]: 4 cycles
- reading array[128]: 4 cycles
- reading array[192]: 20 cycles
- reading array[256]: 4 cycles
- reading array[288]: 4 cycles
- …

what could cause this difference?

- array[192] not in some cache, but others were

inferring cache accesses (2)

some psuedocode:

```
char array[CACHE_SIZE];
AccessAllOf(array);
*other_address += 1;
TimeAccessingArray();
```

suppose during these accesses I discover that `array[128]` is slower to access

probably because `*other_address` loaded into cache + evicted it

what do we know about `other_address`? (select all that apply)

A. same cache tag B. same cache index C. same cache offset
D. diff. cache tag E. diff. cache index F. diff. cache offset

some complications

caches often use physical, not virtual addresses

- (and need to know about physical address to compare index bits)
- (but can infer physical addresses with measurements/asking OS)
- (often OS allocates contiguous physical addresses esp. w/'large pages')

storing/processing timings evicts things in the cache

- (but can compare timing with/without access of interest to check for this)

processor "pre-fetching" may load things into cache before access is timed

- (but can arrange accesses to avoid triggering prefetcher and make sure to measure with memory barriers)

some L3 caches use a simple hash function to select index instead of index bits

exercise: inferring cache accesses (1)

```
char *array;
array = AllocateAlignedPhysicalMemory(CACHE_SIZE);
LoadIntoCache(array, CACHE_SIZE);
if (mystery) {
    *pointer += 1;
}
if (TimeAccessTo(&array[index]) > THRESHOLD) {
    /* pointer accessed */
}
```

suppose `pointer` is `0x1000188`

and cache (of interest) is direct-mapped, 32768 (2^{15}) byte, 64-byte blocks

what array index should we check?

aside

```
array = AllocateAlignedPhysicalMemory(CACHE_SIZE);
LoadIntoCache(array, CACHE_SIZE);
if (mystery) { *pointer += 1; }
if (TimeAccessTo(&array[index]) > THRESHOLD) {
    /* pointer accessed */
}
```

will this detect when pointer accessed? yes

will this detect if mystery is true? not quite

...because branch prediction could started cache access

exercise: inferring cache accesses (2)

```
char *other_array = ...;
char *array;
array = AllocateAlignedPhysicalMemory(CACHE_SIZE);
LoadIntoCache(array, CACHE_SIZE);
other_array[mystery] += 1;
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (TimeAccessTo(&array[i]) > THRESHOLD) {
        /* found something interesting */
    }
}
```

other_array at 0x200400, and interesting index is i=0x800, then what was mystery?

exercise: inferring cache accesses (2)

```
char *array;
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
LoadIntoCache(array, CACHE_SIZE);
if (mystery) {
    *pointer = 1;
}
if (TimeAccessTo(&array[index1]) > THRESHOLD ||
    TimeAccessTo(&array[index2]) > THRESHOLD) {
    /* pointer accessed */
}
```

pointer is 0x1000188

cache is 2-way, 32768 (2^{15}) byte, 64-byte blocks, ???? replacement

what array indexes should we check?

PRIME+PROBE

name in literature: PRIME + PROBE

PRIME: fill cache (or part of it) with values

do thing that uses cache

PROBE: access those values again and see if it's slow

(one of several ways to measure how cache is used)

coined in attacks on AES encryption

example: AES (1)

from Osvik, Shamir, and Tromer, "Cache Attacks and Countermeasures: the Case of AES" (2004)

early AES implementation used lookup tables

goal: detect index into lookup table

- index depended on key + data being encrypted

tricks they did to make this work

- vary data being encrypted
- subtract average time to look for what changes
- lots of measurements

example: AES (2)

from Osvik, Shamir, and Tromer, “Cache Attacks and Countermeasures: the Case of AES” (2004)

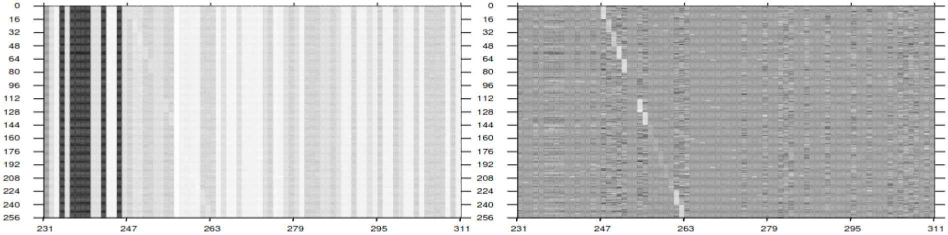


Fig. 5. Prime+Probe attack using 30,000 encryption calls on a 2GHz Athlon 64, attacking Linux 2.6.11 `dm-crypt`. The horizontal axis is the evicted cache set (i.e., $\langle y \rangle$ plus an offset due to the table's location) and the vertical axis is p_0. Left: raw timings (lighter is slower). Right: after subtraction of the average timing of the cache set. The bright diagonal reveals the high nibble of $p_0 = 0x00$.

reading a value

```
char *array;
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
AccessAllOf(array);
other_array[mystery * BLOCK_SIZE] += 1;
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) {
        ...
    }
}
```

with 32KB direct-mapped cache

suppose we find out that `array[0x400]` is slow to access

and other_array starts at address 0x100000

what was mystery?

revisiting an earlier example (1)

```
char *array;
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
LoadIntoCache(array, CACHE_SIZE);
if (mystery) {
    *pointer += 1;
}
if (TimeAccessTo(&array[index]) > THRESHOLD) {
    /* pointer accessed */
}
```

what if mystery is false *but* branch mispredicted?

revisiting an earlier example (2)

cycle #	0	1	2	3	4	5	6	7	8	9	10	11
`movq mystery, %rax`	F	D	R	I	E	E	E	W	C			
`test %rax, %rax`	F	D	R				I	E	W	C		
`jz skip` (mispred.)		F	D	R				I	E	W	C	
`mov pointer, %rax`		F	D	R	I	E	E	E	W			
`mov (%rax), %r8`			F	D	R				I	E	W	
`add $1, %r8`			F	D	R							
`mov %r8, %rax`				F	D	R						
…												
`skip: ...`								F	D	R		

avoiding/triggering this problem

```
if (something false) {
    access *pointer;
}
```

what can we do to make access more/less likely to happen?

reading a value without really reading it

```
char *array;
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
AccessAllOf(array);
if (something false) {
    other_array[mystery * BLOCK_SIZE] += 1;
}
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) {
        ...
    }
}
```

if branch mispredicted, cache access may still happen

can find the value of `mystery`

seeing past a segfault? (1)

```
Prime();
if (something false) {
    triggerSegfault();
    Use(*pointer);
}
Probe();
```

could cache access for `*pointer` still happen?

yes, if:

- branch for if statement mispredicted, and
- `*pointer` starts before segfault detected

seeing past a segfault? (2)

operations in virtual memory lookup:

- translate virtual to physical address
- check if access is permitted by permission bits

Intel processors: looks like these were separate steps, so...

```
Prime();
if (something false) {
    int value = ReadMemoryMarkedNonReadableInPageTable();
    access other_array[value * ...];
}
Probe();
```

seeing past a segfault? (2)

operations in virtual memory lookup:

- translate virtual to physical address
- check if access is permitted by permission bits

Intel processors: looks like these were separate steps, so...

```
Prime();
if (something false) {
    int value = ReadMemoryMarkedNonReadableInPageTable();
    access other_array[value * ...];
}
Probe();
```

seeing past a segfault? (2)

operations in virtual memory lookup:

- translate virtual to physical address
- check if access is permitted by permission bits

Intel processors: looks like these were separate steps, so...

```
Prime();
if (something false) {
    int value = ReadMemoryMarkedNonReadableInPageTable();
    access other_array[value * ...];
}
Probe();
```

seeing past a segfault? (2)

operations in virtual memory lookup:

- translate virtual to physical address
- check if access is permitted by permission bits

Intel processors: looks like these were separate steps, so...

```
Prime();
if (something false) {
    int value = ReadMemoryMarkedNonReadableInPageTable();
    access other_array[value * ...];
}
Probe();
```

Meltdown

from Lipp et al, “Meltdown: Reading Kernel Memory from User Space”

```
    // %rcx = kernel address
    // %rbx = array to load from to cause eviction
    xor %rax, %rax        // rax <- 0
retry:
    // rax <- memory[kernel address] (segfaults)
        // but check for segfault done out-of-order on Intel
    movb (%rcx), %al
    // rax <- memory[kernel address] * 4096 [speculated]
    shl $0xC, %rax
    jz retry                  // not-taken branch
    // access array[memory[kernel address] * 4096]
    mov (%rbx, %rax), %rbx
```

Meltdown

from Lipp et al, “Meltdown: Reading Kernel Memory from User Space”

```
    // %rcx = kernel address
    // %rbx = ar                              viction
    xor %rax, %ra
retry:
    // rax <- me                              ults)
        // but check for segfault done out-of-order on Intel
    movb (%rcx), %al
    // rax <- memory[kernel address] * 4096 [speculated]
    shl $0xC, %rax
    jz retry                    // not-taken branch
    // access array[memory[kernel address] * 4096]
    mov (%rbx, %rax), %rbx
```

space out accesses by 4096
ensure separate cache sets and
avoid triggering prefetcher

Meltdown

from Lipp et al, “Meltdown: Reading Kernel Memory from User Space”

```
    // %rcx = kernel address
    // %rbx                                                  on
    xor %rax
retry:
    // rax <
        // but check for segfault done out-of-order on Intel
    movb (%rcx), %al
    // rax <- memory[kernel address] * 4096 [speculated]
    shl $0xC, %rax
    jz retry                    // not-taken branch
    // access array[memory[kernel address] * 4096]
    mov (%rbx, %rax), %rbx
```

repeat access if zero
apparently value of zero speculatively read
when real value not yet available

Meltdown

from Lipp et al, "Meltdown: Reading Kernel Memory from User Space"

```
    // %rcx = kernel address
    // %rbx =                                       ion
    xor %rax
retry:
    // rax <-                                       )
        // but check for segfault done out-of-order on Intel
    movb (%rcx), %al
    // rax <- memory[kernel address] * 4096 [speculated]
    shl $0xC, %rax
    jz retry                      // not-taken branch
    // access array[memory[kernel address] * 4096]
    mov (%rbx, %rax), %rbx
```

access cache to allow measurement later in paper with FLUSH+RELOAD instead of PRIME+PROBE technique

Meltdown

from Lipp et al, "Meltdown: Reading Kernel Memory from User Space"

```
// %rcx = kernel address
```

segfault actually happens eventually

option 1: okay, just start a new process every time

option 2: way of suppressing exception (transactional memory support)

```
    // but check for segfault done out-of-order on Intel
movb (%rcx), %al
// rax <- memory[kernel address] * 4096 [speculated]
shl $0xC, %rax
jz retry                    // not-taken branch
// access array[memory[kernel address] * 4096]
mov (%rbx, %rax), %rbx
```

Meltdown fix

HW: permissions check done with/before physical address lookup
- was already done by AMD, ARM apparently?
- now done by Intel

SW: separate page tables for kernel and user space
- don't have sensitive kernel memory pointed to by page table
- when user-mode code running
- unfortunate performance problem
- exceptions start with code that switches page tables

reading a value without really reading it

```
char *array;
posix_memalign(&array, CACHE_SIZE, CACHE_SIZE);
AccessAllOf(array);
if (something false) {
    other_array[mystery * BLOCK_SIZE] += 1;
}
for (int i = 0; i < CACHE_SIZE; i += BLOCK_SIZE) {
    if (CheckIfSlowToAccess(&array[i])) {
        ...
    }
}
```

if branch mispredicted, cache access may still happen

can find the value of `mystery`

mistraining branch predictor?

```
if (something) {
    CodeToRunSpeculatively()
}
```

how can we have ‘something’ be false, but predicted as true

run lots of times with something true

then do actually run with something false

contrived(?) vulnerable code (1)

suppose this C code is run with extra privileges

(e.g. in system call handler, library called from JavaScript in webpage, etc.)

assume `x` chosen by attacker

(example from original Spectre paper)

```
if (x < array1_size)
        y = array2[array1[x] * 4096];
```

the out-of-bounds access (1)

```
char array1[...];
...
int secret;
...
y = array2[array1[x] * 4096];
```

suppose array1 is at `0x1000000` and

secret is at `0x103F0003`;

what x do we choose to make `array1[x]` access first byte of secret?

the out-of-bounds access (2)

```
unsigned char array1[...];
...
int secret;
...
y = array2[array1[x] * 4096];
```

suppose our cache has 64-byte blocks and 8192 sets

and `array2[0]` is stored in cache set 0

if the above evicts something in cache set 128,
then what do we know about `array1[x]`?

the out-of-bounds access (2)

```
unsigned char array1[...];
...
int secret;
...
y = array2[array1[x] * 4096];
```

suppose our cache has 64-byte blocks and 8192 sets

and `array2[0]` is stored in cache set 0

if the above evicts something in cache set 128,
then what do we know about `array1[x]`?

is 2 or 130

exploit with contrived(?) code

```
/* in kernel: */
int systemCallHandler(int x) {
    if (x < array1_size)
        y = array2[array1[x] * 4096];
    return y;
}
```

```
/* exploiting code */
    /* step 1: mistrain branch predictor */
for (a lot) {
    systemCallHandler(0 /* less than array1_size */);
}
    /* step 2: evict from cache using misprediction */
Prime();
systemCallHandler(targetAddress - array1Address);
int evictedSet = ProbeAndFindEviction();
int targetValue = (evictedSet - array2StartSet) / setsPer4K;
```

really contrived?

```
char *array1; char *array2;
if (x < array1_size)
    y = array2[array1[x] * 4096];
```

times 4096 shifts so we can get lower bits of target value

so all bits effect what cache block is used

really contrived?

```
char *array1; char *array2;
if (x < array1_size)
    y = array2[array1[x] * 4096];
```

times 4096 shifts so we can get lower bits of target value

so all bits effect what cache block is used

```
int *array1; int *array2;
if (x < array1_size)
    y = array2[array1[x]];
```

will still get *upper* bits of array1[x] (can tell from cache set)

can still read arbitrary memory!

want memory at 0x10000?

upper bits of 4-byte integer at 0x0FFFE

bounds check in kernel

```
if (x < array1_size) {
    y = array2[array1[x]];
}
```

our template

```
void SomeSystemCallHandler(int index) {
    if (index > some_table_size)
        return ERROR;
    int kind = table[index];
    switch (other_table[kind].foo) {
        ...
    }
}
```

actual code

bounds check in kernel

```
if (x < array1_size) {
    y = array2[array1[x]];
}
```

our template

```
void SomeSystemCallHandler(int index) {
    if (index > some_table_size)
        return ERROR;
    int kind = table[index];
    switch (other_table[kind].foo) {
        ...
    }
}
```

actual code

bounds check in kernel

```
if (x < array1_size) {
    y = array2[array1[x]];
}
```

our template

```
void SomeSystemCallHandler(int index) {
    if (index > some_table_size)
        return ERROR;
    int kind = table[index];
    switch (other_table[kind].foo) {
        ...
    }
}
```

actual code

bounds check in kernel

```
if (x < array1_size) {
    y = array2[array1[x]];
}
```

our template

```
void SomeSystemCallHandler(int index) {
    if (index > some_table_size)
        return ERROR;
    int kind = table[index];
    switch (other_table[kind].foo) {
        ...
    }
}
```

actual code

privilege levels?

vulnerable code runs with higher privileges

so far: higher privileges = kernel mode

but other common cases of higher privileges

example: scripts in web browsers

JavaScript

JavaScript: scripts in webpages

not supposed to be able to read arbitrary memory, but...

can access arrays to examine caches

and could take advantage of some browser function being vulnerable

JavaScript

JavaScript: scripts in webpages

not supposed to be able to read arbitrary memory, but...

can access arrays to examine caches

and could take advantage of some browser function being vulnerable

or — doesn't even need browser to supply vulnerable code itself!

just-in-time compilation?

for performance, compiled to machine code, run in browser

not supposed to be access arbitrary browser memory

example JavaScript code from paper:

```
if (index < simpleByteArray.length) {
    index = simpleByteArray[index | 0];
    index = (((index * 4096)|0) & (32*1024*1024-1))|0;
    localJunk ^= probeTable[index|0]|0;
}
```

web page runs a lot to train branch predictor

then does run with out-of-bounds index

examines what's evicted by probeTable access

supplying own attack code?

JavaScript: could supply own attack code

turns out also possible with kernel mode scenario

trick: don't need to *actually run* code

...just need branch predictor to fetch it!

other misprediction

so far: talking about mispredicting direction of branch

what about mispredicting target of branch in, e.g.:

```
// possibly from C code like:
//    (*function_pointer)();
jmp *%rax

// possibly from C code like:
//         switch(rcx) { ... }
jmp *(%rax,%rcx,8)
```

an idea for predicting indirect jumps

for jmps like `jmp *%rax` predict target with cache:

bottom 12 bits of jmp address	last seen target
0x0–0x7	0x200000
0x8–0xF	0x440004
0x10-0x18	0x4CD894
0x18-0x20	0x510194
0x20-0x28	0x4FF194
…	…
0xFF8–0xFFF	0x3F8403

Intel Haswell CPU did something similar to this

uses bits of last several jumps, not just last one

can mistrain this branch predictor

using mispredicted jump

1: find some kernel function with `jmp *%rax`

2: mistrain branch target predictor for it to jump to chosen code

  use code at address that conflicts in “recent jumps cache”

3: have chosen code be attack code (e.g. array access)

  either write special code OR

  find suitable instructions (e.g. array access) in existing kernel code

Spectre variants

- showed Spectre variant 1 (array bounds), 2 (indirect jump)
  - from original paper

- other possible variations:
  - could cause other things to be mispredicted
    - prediction of where functions return to?
    - values instead of which code is executed?
  - could use side-channel other than data cache changes
    - instruction cache
    - cache of pending stores not yet committed
    - contention for resources on multi-threaded CPU core
    - branch prediction changes
    - …

some Linux kernel mitigations (1)

replace `array[x]` with
`array[x & ComputeMask(x, size)]`

...where ComputeMask() returns

- 0 if x $>$ size
- `0xFFFF..F` if x $\leq$ size

...and ComputeMask() does not use jumps:

```
mov x, %r8
mov size, %r9
cmp %r9, %r8
sbb %rax, %rax  // sbb = subtract with borrow
    // either 0 or -1
```

some Linux kernel mitigations (2)

for indirect branches:

with hardware help:

- separate indirect (computed) branch prediction for kernel v user mode
- other branch predictor changes to isolate better

without hardware help:

- transform `jmp *(%rax)`, etc. into code that will only predicted to jump to safe locations (by writing assembly very carefully)

only safe prediction

as replacement for `jmp *(%rax)`

code from Intel's "Retpoline: A Branch Target Injection Mitigation"

```
        call load_label
    capture_ret_spec:    /* <-- want prediction to go here */
        pause
        lfence
        jmp capture_ret_spec
    load_label:
        mov %rax, (%rsp)
        ret
```

backup slides

backup slides

beyond 1-bit predictor

devote *more space* to storing history

main goal: rare exceptions don't immediately change prediction

example: branch taken 99% of the time

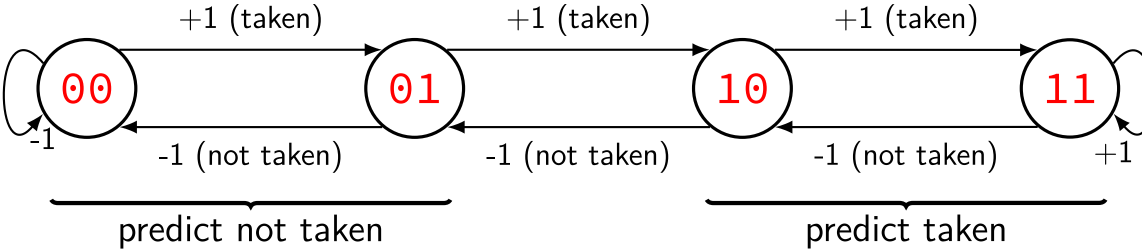
1-bit predictor: wrong about 2% of the time

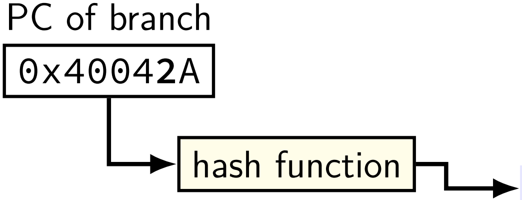
- 1% when branch not taken
- 1% of taken branches right after branch not taken

new predictor: wrong about 1% of the time

- 1% when branch not taken

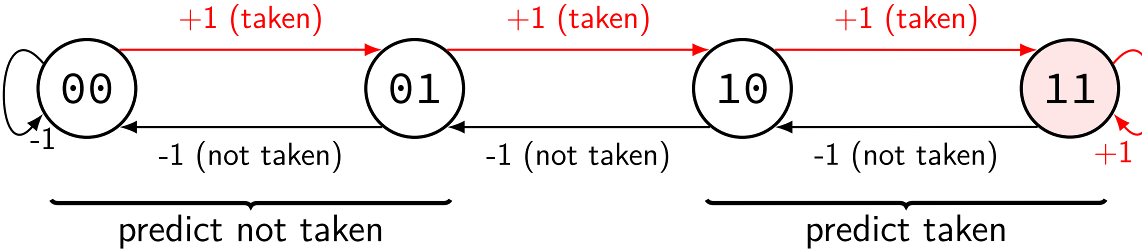
2-bit saturating counter





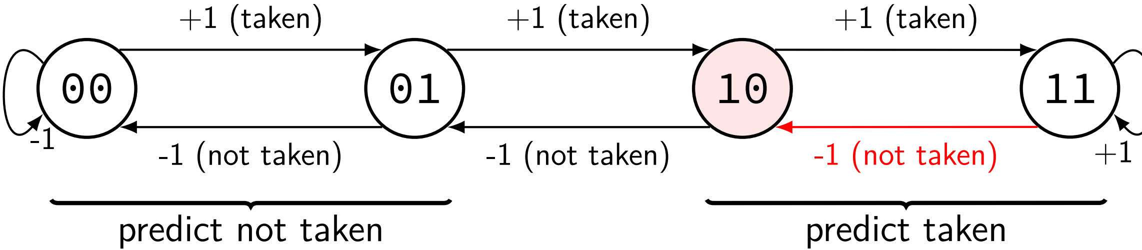
index	*counter*
0	11
1	01
2	11
...	...
14	10
15	00

2-bit saturating counter



branch always taken:
value increases to 'strongest' taken value

2-bit saturating counter



branch almost always taken, then not taken once:
still predicted as taken

example

`0x40041B`	`movq $4, %rax`
`0x400423`	`· · ·`
`0x400429`	`decq %rax`
`0x40042A`	`jz 0x400423`
`0x40042B`	`· · ·`

iter.	table before	prediction	outcome	table after
1	01	not taken	taken	10
2	10	taken	taken	11
3	11	taken	taken	11
4	11	taken	not taken	10
1	10	taken	taken	11
2	11	taken	taken	11
3	11	taken	taken	11
4	11	taken	not taken	10
1	10	taken	taken	11
…	…	…	…	…

generalizing saturating counters

2-bit counter: ignore one exception to taken/not taken

3-bit counter: ignore more exceptions

`000` $\leftrightarrow$ `001` $\leftrightarrow$ `010` $\leftrightarrow$ `011` $\leftrightarrow$ `100` $\leftrightarrow$ `101` $\leftrightarrow$ `110` $\leftrightarrow$ `111`

`000`-`011`: not taken

`100`-`111`: taken

exercise

use 2-bit predictor on this loop

executed in outer loop (not shown) many, many times

what is the conditional branch misprediction rate?

```
int i = 0;
while (true) {
  if (i % 3 == 0) goto next;
  ...
next:
  i += 1;
  if (i == 50) break;
}
```

branch patterns

```
i = 4;
do {
    ...
    i -= 1;
} while (i != 0);
```

typical pattern for jump to top of do-while above:

TTTN TTTN TTTN TTTN TTTN...(T = taken, N = not taken)

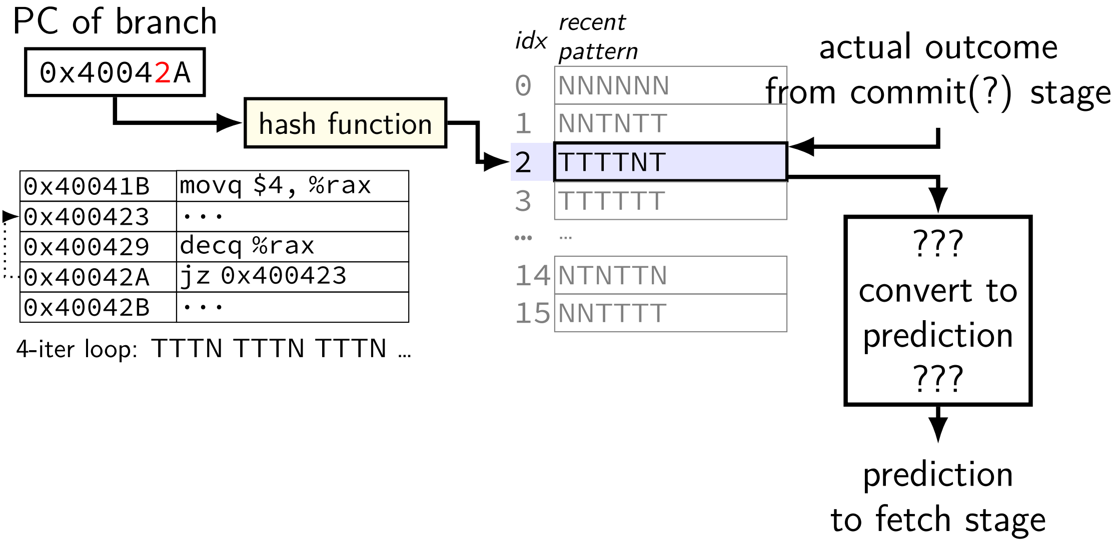
goal: take advantage of recent pattern to make predictions

just saw 'NTTTNT'? predict T next

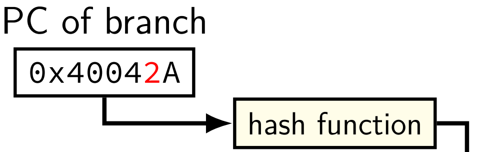
'TNTTTN'? predict T; 'TTNTTT'? predict N next

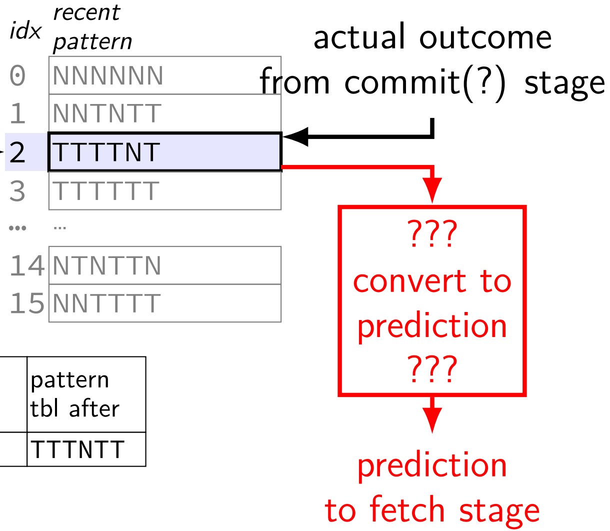
...

local pattern predictor (incomplete)



local pattern predictor (incomplete)



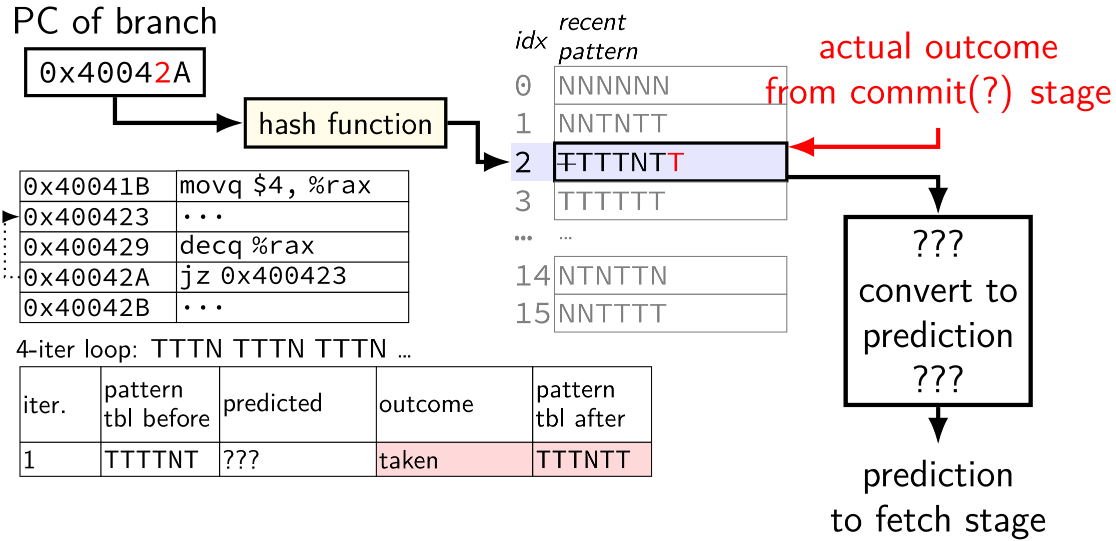


0x40041B	movq $4, %rax
0x400423	···
0x400429	decq %rax
0x40042A	jz 0x400423
0x40042B	···

4-iter loop: TTTN TTTN TTTN ...

iter.	pattern tbl before	predicted	outcome	pattern tbl after
1	TTTTNT	???	taken	TTTNTT

local pattern predictor (incomplete)

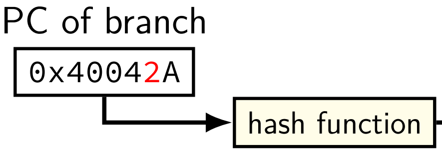


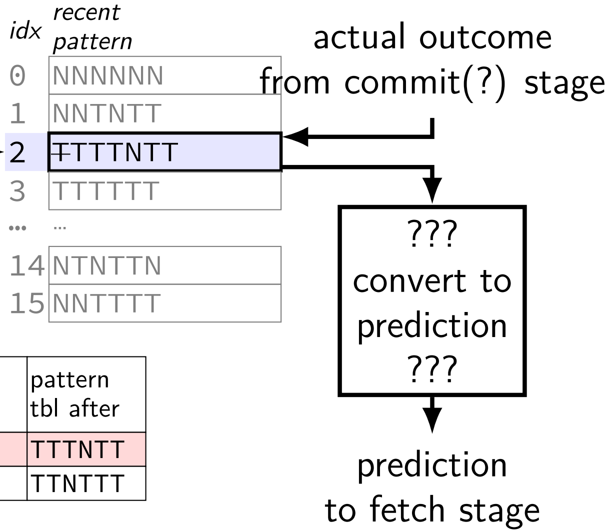
0x40041B	movq $4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jz 0x400423
0x40042B	...

4-iter loop: TTTN TTTN TTTN ...

iter.	pattern tbl before	predicted	outcome	pattern tbl after
1	TTTTNT	???	taken	TTTNTT

local pattern predictor (incomplete)





0x40041B	movq $4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jz 0x400423
0x40042B	...

4-iter loop: TTTN TTTN TTTN ...

iter.	pattern tbl before	predicted	outcome	pattern tbl after
1	TTTTNT	???	taken	TTTNTT
2	TTTNTT	???	taken	TTNTTT

local pattern predictor (incomplete)

PC of branch

0x40042A

hash function

idx	*recent pattern*
0	NNNNNN
1	NNTNTT
2	~~TT~~TTNTTT
3	TTTTTT
…	…
14	NTNTTN
15	NNTTTT

actual outcome from commit(?) stage

???
convert to prediction
???

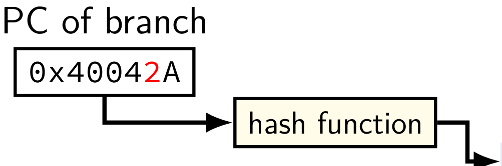
prediction to fetch stage

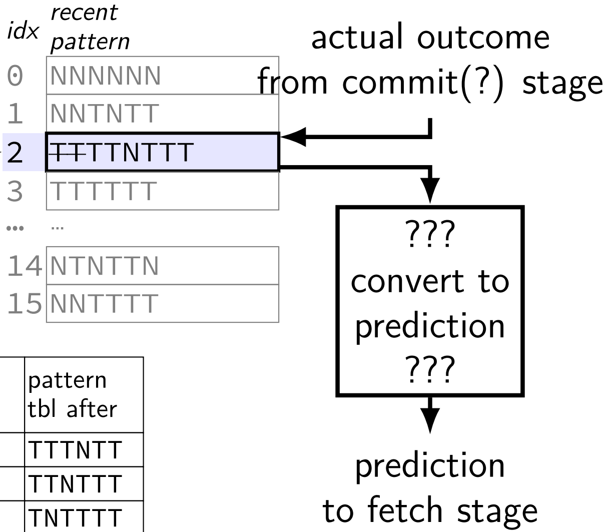
0x40041B	movq $4, %rax
0x400423	…
0x400429	decq %rax
0x40042A	jz 0x400423
0x40042B	…

4-iter loop: TTTN TTTN TTTN …

iter.	pattern tbl before	predicted	outcome	pattern tbl after
1	TTTTNT	???	taken	TTTNTT
2	TTTNTT	???	taken	TTNTTT
3	TTNTTT	???	taken	TNTTTT

local pattern predictor (incomplete)





0x40041B	movq $4, %rax
0x400423	...
0x400429	decq %rax
0x40042A	jz 0x400423
0x40042B	...

4-iter loop: TTTN TTTN TTTN ...

iter.	pattern tbl before	predicted	outcome	pattern tbl after
1	TTTTNT	???	taken	TTTNTT
2	TTTNTT	???	taken	TTNTTT
3	TTNTTT	???	taken	TNTTTT
4	TNTTTT	???	not taken	NTTTTN
1	NTTTTN	???	taken	TTTTNT
2	TTTTNT	???	taken	TTTNTT

recent pattern to prediction?

easy cases:

just saw TTTTTT: predict T

just saw NNNNNN: predict N

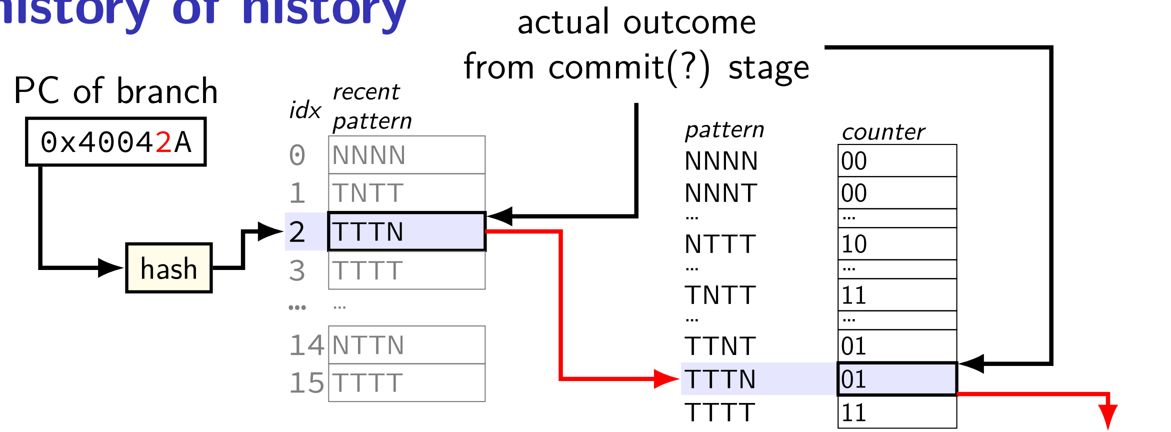
just saw TNTNTN: predict T

hard cases:

TTNTTTT

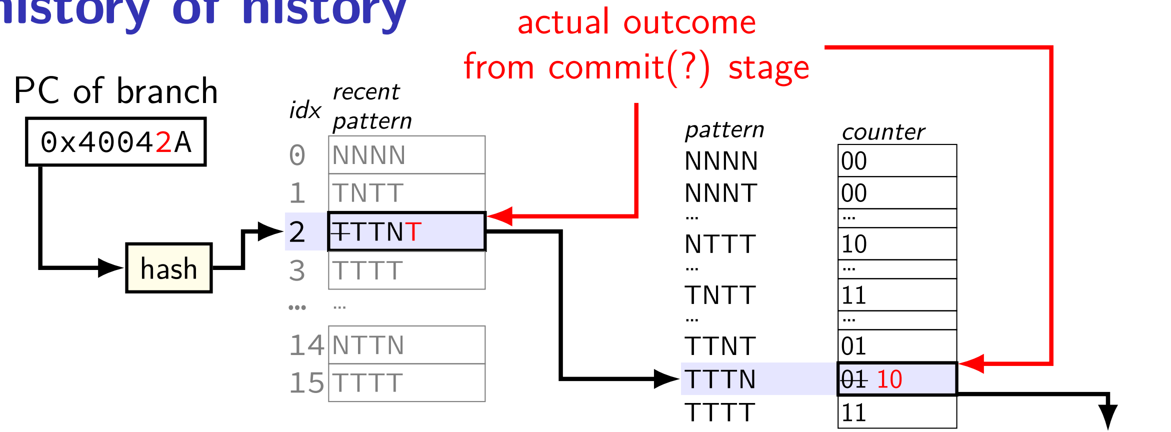
- predict T? loop with many iterations (NTTTTTTTNTTTTTTTNTTTTTT...)
- predict T? if statement mostly taken (TTTTNTTNTTTTTTTTTTTNTTTT...)
- predict N? loop with 5 iterations (NTTTTNTTTTNTTTTNTTTTNTT...)

history of history



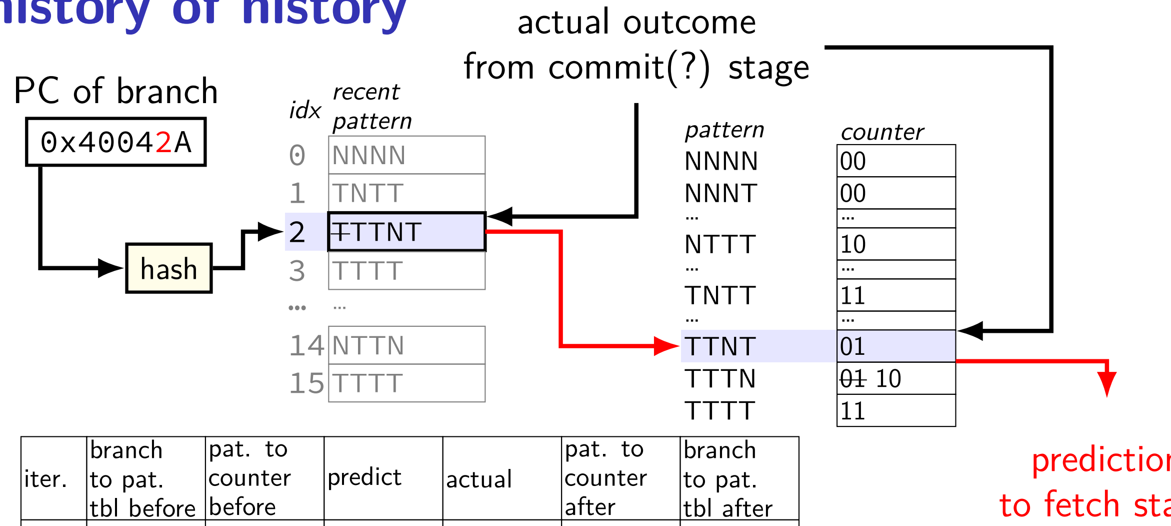
iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT

history of history



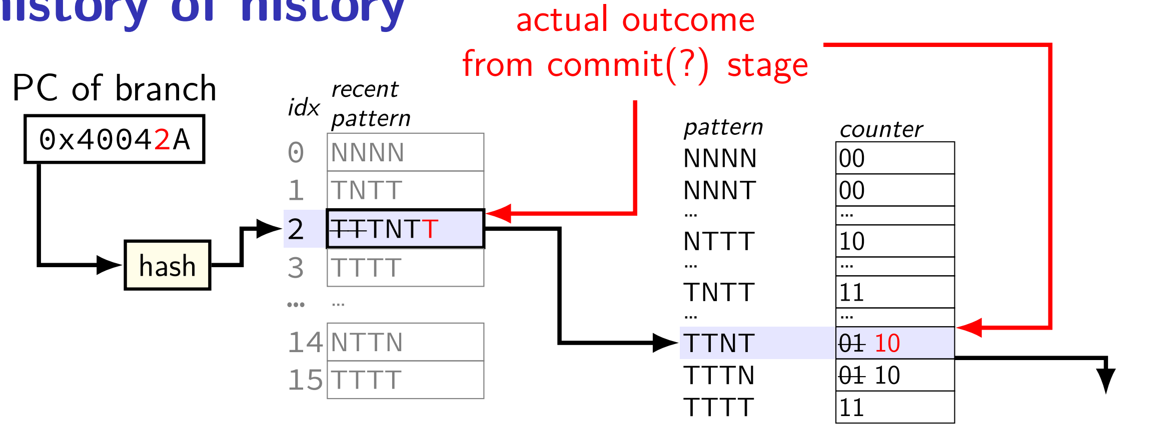
iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT

history of history



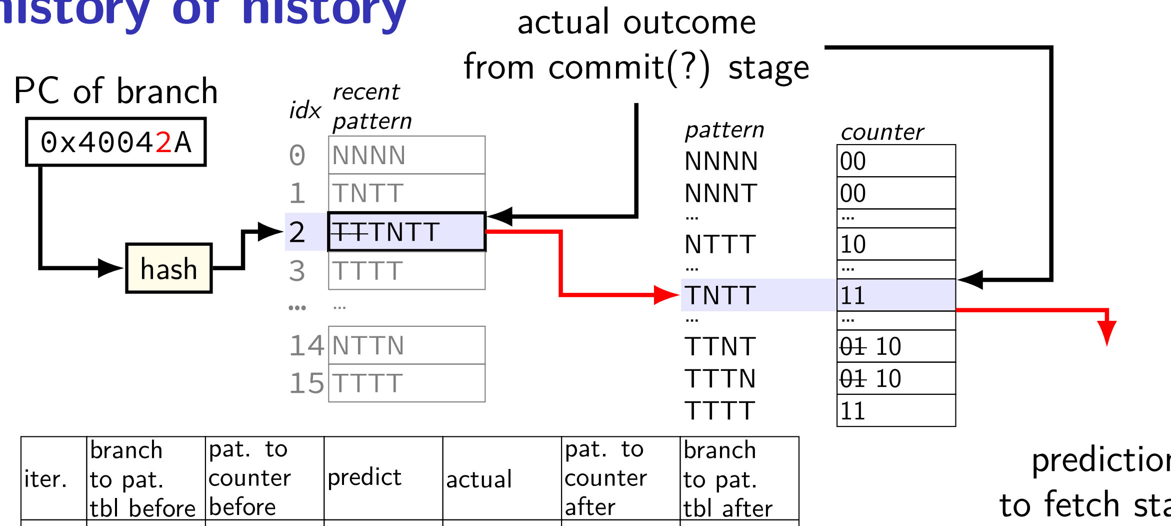
iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT

history of history



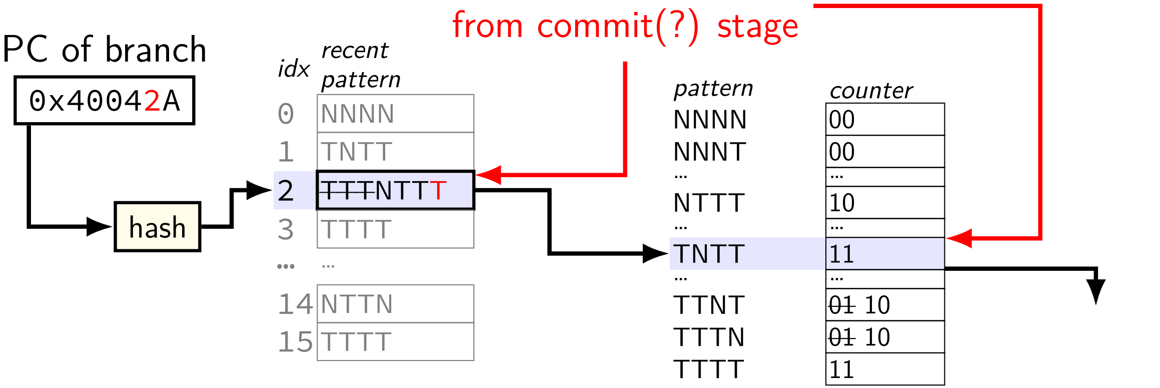
iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT

history of history



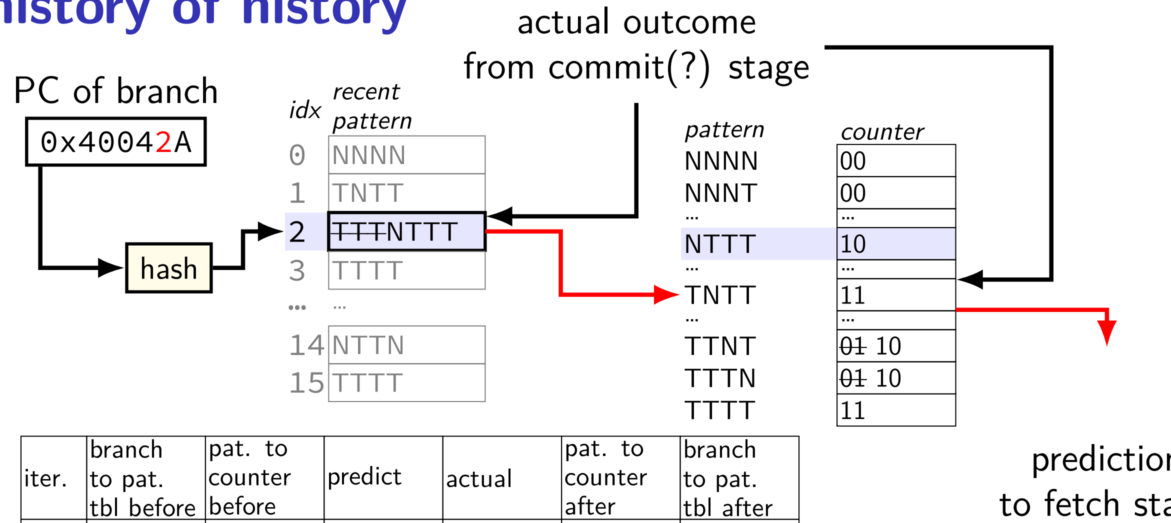
iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT
3	TNTT	11	taken	taken	11	NTTT

history of history



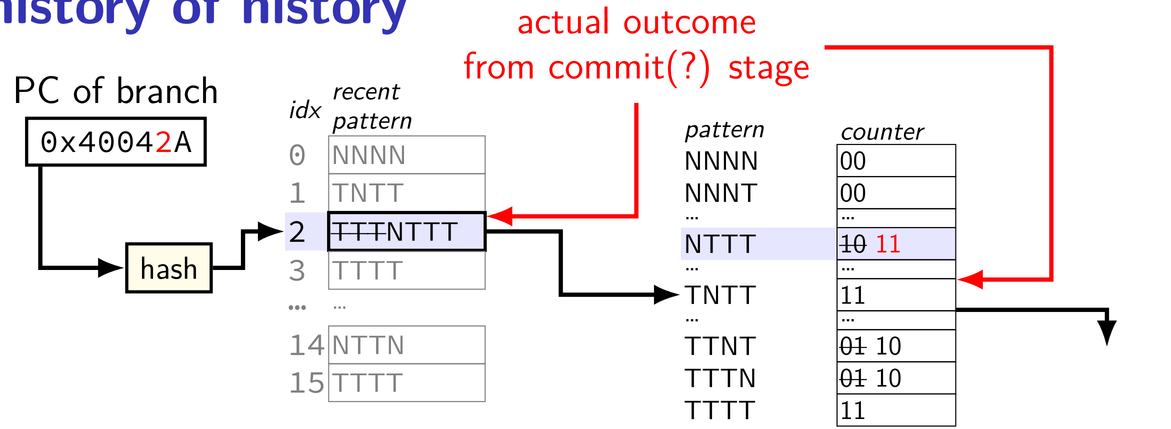
iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT
3	TNTT	11	taken	taken	11	NTTT

history of history



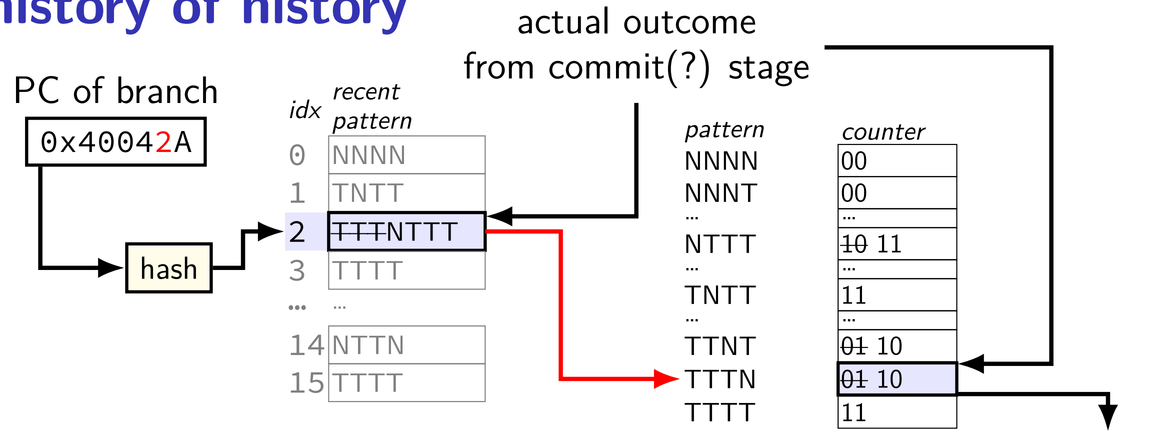
iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT
3	TNTT	11	taken	taken	11	NTTT
4	NTTT	01	not taken	taken	10	TTTT

history of history



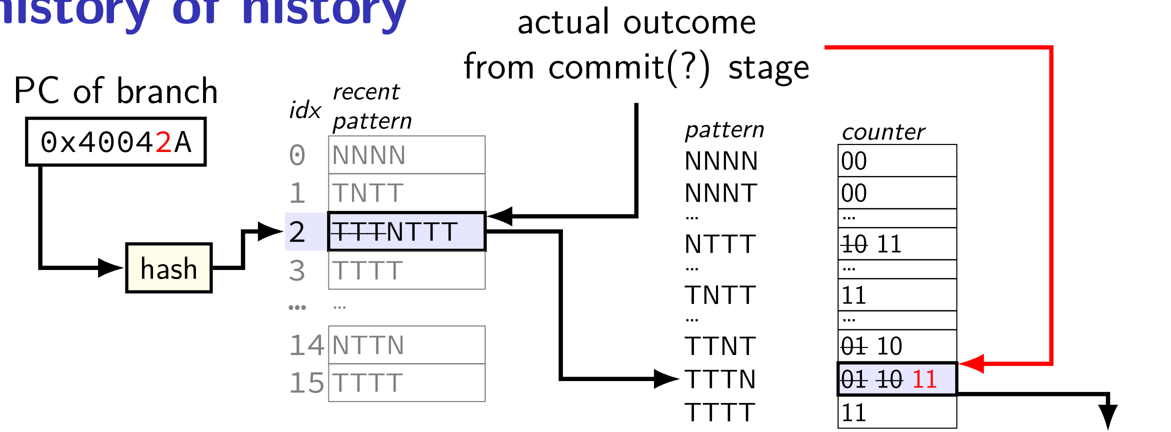
iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT
3	TNTT	11	taken	taken	11	NTTT
4	NTTT	01	not taken	taken	10	TTTT

history of history



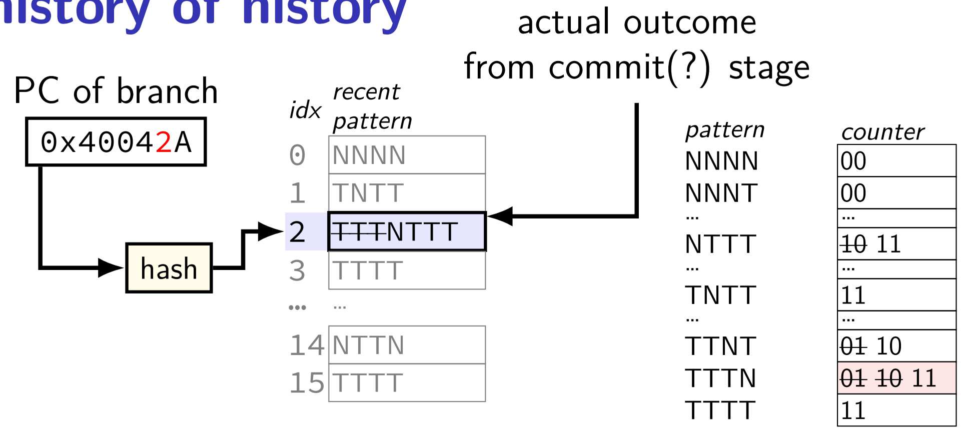
iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT
3	TNTT	11	taken	taken	11	NTTT
4	NTTT	01	not taken	taken	10	TTTT
1	TTTN	10	taken	taken	11	TTNT

history of history



iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT
3	TNTT	11	taken	taken	11	NTTT
4	NTTT	01	not taken	taken	10	TTTT
1	TTTN	10	taken	taken	11	TTNT

history of history



iter.	branch to pat. tbl before	pat. to counter before	predict	actual	pat. to counter after	branch to pat. tbl after
1	TTTN	01	not taken	taken	10	TTNT
2	TTNT	01	not taken	taken	10	TNTT
3	TNTT	11	taken	taken	11	NTTT
4	NTTT	01	not taken	taken	10	TTTT
1	TTTN	10	taken	taken	11	TTNT

prediction
to fetch sta

local patterns and collisions (1)

```
i = 10000;
do {
    p = malloc(...);
    if (p == NULL) goto error;  // BRANCH 1
    ...
} while (i-- != 0); // BRANCH 2
```

what if branch 1 and branch 2 hash to same table entry?

local patterns and collisions (1)

```
i = 10000;
do {
    p = malloc(...);
    if (p == NULL) goto error;  // BRANCH 1
    ...
} while (i-- != 0); // BRANCH 2
```

---

what if branch 1 and branch 2 hash to same table entry?

pattern: TNTNTNTNTNTNTNTNT...

actually no problem to predict!

local patterns and collisions (2)

```
i = 10000;
do {
    if (i % 2 == 0) goto skip; // BRANCH 1
    ...
    p = malloc(...);
    if (p == NULL) goto error;  // BRANCH 2
skip: ...
} while (i-- != 0); // BRANCH 3
```

---

what if branch 1 and branch 2 and branch 3 hash to same table entry?

local patterns and collisions (2)

```
i = 10000;
do {
    if (i % 2 == 0) goto skip; // BRANCH 1
    ...
    p = malloc(...);
    if (p == NULL) goto error;  // BRANCH 2
skip: ...
} while (i-- != 0); // BRANCH 3
```

---

what if branch 1 and branch 2 and branch 3 hash to same table entry?

pattern: TTNNTTNNTTNNTTNNTT

also no problem to predict!

local patterns and collisions (3)

```
i = 10000;
do {
    if (A) goto one  // BRANCH 1
    ...
one:
    if (B) goto two  // BRANCH 2
    ...
two:
    if (A or B) goto three // BRANCH 3
    ...
    if (A and B) goto three // BRANCH 4
    ...
three:
    ... // changes A, B
} while (i-- != 0);
```

---

what if branch 1-4 hash to same table entry?

better for prediction of branch 3 and 4

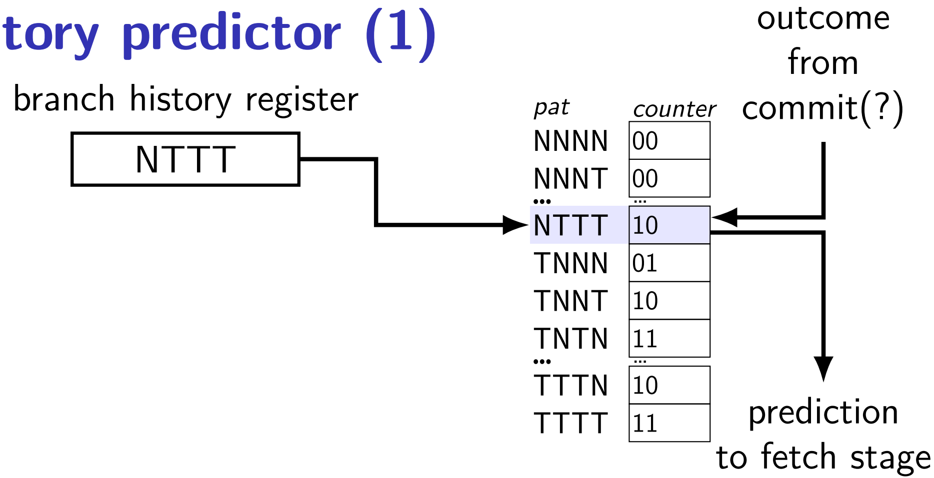
global history predictor: idea

one predictor idea: ignore the PC

just record taken/not-taken pattern for all branches

lookup in big table like for local patterns

global history predictor (1)



global history predictor (1)

```
i = 10000;
do {
  if (i % 2 == 0) goto skip;
  ...
  if (p == NULL) goto error;
skip:
  ...
} while (i-- != 0);
```

branch history register

NTTT

outcome from commit(?)

pat	*counter*
NNNN	00
NNNT	00
...	...
NTTT	10
TNNN	01
TNNT	10
TNTN	11
...	...
TTTN	10
TTTT	11

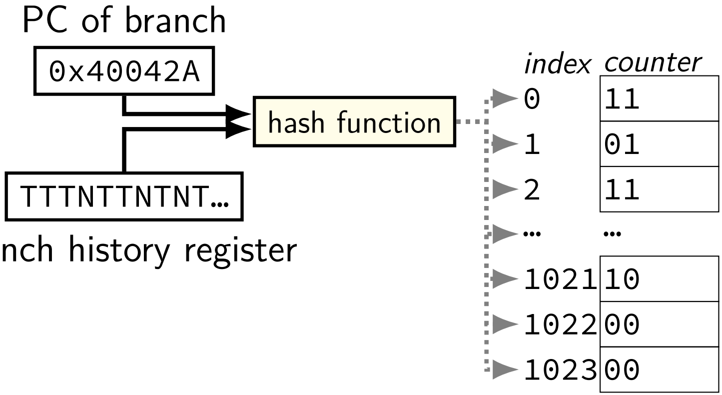
prediction to fetch stage

iter./ branch	history before	counter before	predict	outcome	counter after	history after
0/mod 2	NTTT	10	taken	taken	11	TTTT
0/loop	TTTT			taken		TTTT
1/mod 2	TTTT			not taken		TTTN
1/error	TTTN			not taken		TTNN
1/loop	TNNT			taken		NNTT
2/mod 2	NNTT			taken		NTTT
2/loop	TTTT			taken		TTTT

correlating predictor

global history *and* local info good together

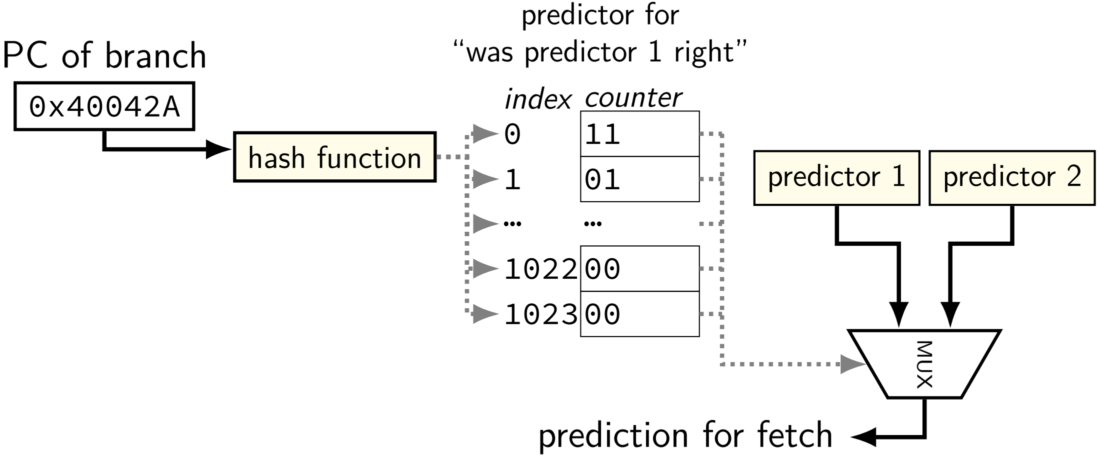
one idea: combine history register + PC ("gshare")



mixing predictors

different predictors good at different times

one idea: have two predictors, + predictor to predict which is right



loop count predictors (1)

```
for (int i = 0; i < 64; ++i)
    ...
```

can we predict this perfectly with predictors we've seen

yes — local or global history with 64 entries

but this is very important — more efficient way?

loop count predictors (2)

loop count predictor idea: look for NNNNNNT+repeat (or TTTTTTN+repeat)

track for each possible loop branch:

- how many repeated Ns (or Ts) so far
- how many repeated Ns (or Ts) last time before one T (or N)
- something to indicate this pattern is useful?

known to be used on Intel

benchmark results

from 1993 paper

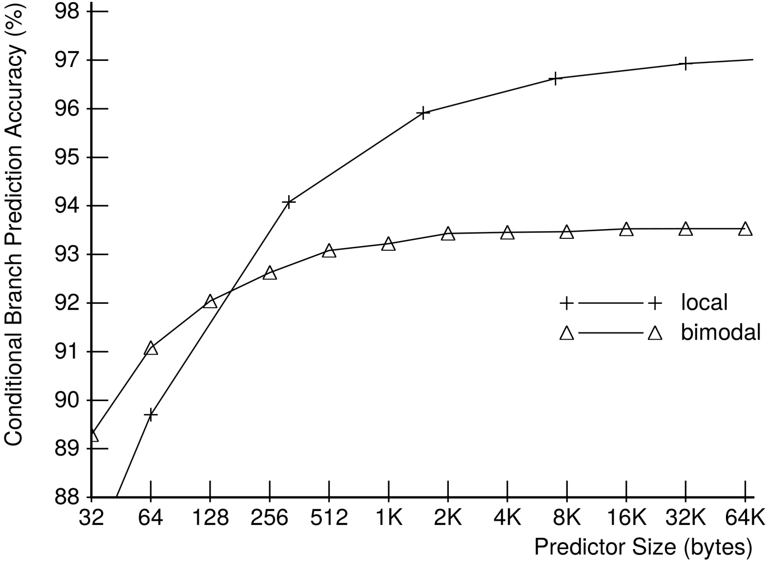
(not representative of modern workloads?)

rate for conditional branches on benchmark

variable table sizes

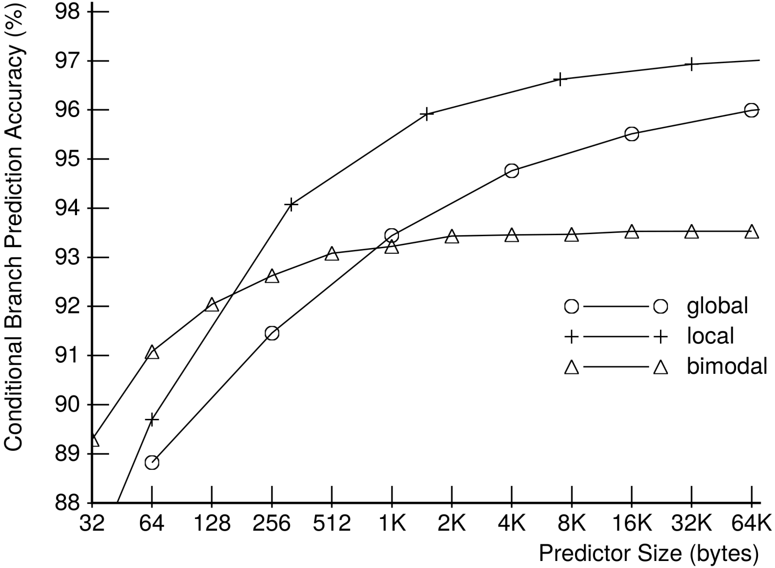
2-bit ctr + local history

from McFarling, "Combining Branch Predictors" (1993)



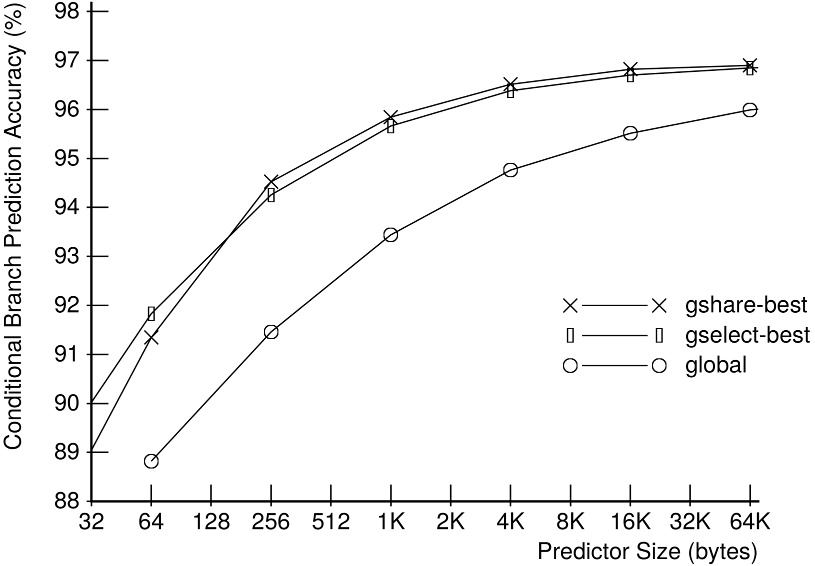
2-bit (bimodal) + local + global hist

from McFarling, “Combining Branch Predictors” (1993)



global + hash(global+PC) (gshare/gselect)

from McFarling, “Combining Branch Predictors” (1993)



real BP?

details of modern CPU's branch predictors often not public

but...

Google Project Zero blog post with reverse engineered details

https://googleprojectzero.blogspot.com/2018/01/reading-privileged-memory-with-side.html

for RE'd BTB size:

https://xania.org/201602/haswell-and-ivy-btb

reverse engineering Haswell BPs

branch target buffer

- 4-way, 4096 entries
- ignores bottom 4 bits of PC?
- hashes PC to index by shifting + XOR
- seems to store 32 bit offset from PC (not all 48+ bits of virtual addr)

indirect branch predictor

- like the global history + PC predictor we showed, but...
- uses history of recent branch addresses instead of taken/not taken
- keeps some info about last 29 branches

what about conditional branches??? loops???

- couldn't find a reasonable source